



POLITECNICO
MILANO 1863

MASTER DEGREE IN MUSIC AND ACOUSTIC ENGINEERING
NUMERICAL MODELING AND SIMULATION IN ACOUSTIC

NMSA HWs

REPORT OF THE HOMEWORKS

STUDENT

Giacomo De Toni, Matéo Vitalone
Student IDs: 273703, 273344

ADVISOR

Prof. Ilario Mazzieri

SUMMARY

1	Homework 1	1
1.1	Weak Form and FE Discretization	1
1.1.1	Strong Formulation	1
1.1.2	Weak Formulation	1
1.1.3	Finite Element Discretization and Algebraic System	2
1.2	MATLAB Solver	3
1.2.1	Verification Test Setup	3
1.2.2	Numerical Results	4
1.2.3	Convergence Analysis	5
1.3	Wave Reflection at an Impedance Jump	7
1.3.1	Problem Setup	7
1.3.2	Mechanical Impedance of the Two Media	7
1.3.3	Reflection Coefficient at the Interface	7
1.3.4	Expected Qualitative Behavior in Amplitude and Phase	8
1.3.5	Numerical Results	8
1.3.6	Physical Interpretation	8
1.4	Numerical Dispersion Analysis	9
1.4.1	Eigenvalue Problem Setting	10
1.4.2	Continuous and Discrete Formulations	10
1.4.3	Dispersion Curve and Numerical Results	10
1.4.4	Physical Interpretation	10
1.5	Perfectly Matched Layer (PML)	12
1.5.1	Damping Profile	12
1.5.2	Complex Stretching and Modified Coefficients	12
1.5.3	Impedance Matching and Exponential Decay	12
1.5.4	Boundary Conditions on the Extended Domain	13
1.5.5	Numerical Results and Comparison	13
1.5.6	Physical Interpretation	13
2	Homework 2	15
2.1	Equivalence to the Second-Order Wave Equation	15
2.1.1	Strong Formulation	15
2.1.2	Decoupling the System	15
2.1.3	Determination of Suitable Boundary Conditions	16
2.1.4	Equivalence	16
2.2	Conservation of Energy	16
2.2.1	Definition of the Energy Functional	16
2.2.2	Time Derivative of the Energy	16
2.2.3	Substitution of the Strong Formulation	17
2.2.4	Algebraic Simplification and the Product Rule	17

2.2.5	Application of the Fundamental Theorem of Calculus	17
2.2.6	Application of Boundary Conditions	17
2.2.7	Energy Conservation	17
2.3	Weak Form and Spectral Element Discretization	18
2.3.1	Function Spaces	18
2.3.2	Weak Formulation	18
2.3.3	Spectral Element Discretization	19
2.3.4	Galerkin System and Matrix Assembly	19
2.3.5	Final Block System	19
2.4	MATLAB Solver	20
2.4.1	Numerical Setup and Time Integration	20
2.4.2	Wave Propagation Snapshots	21
2.4.3	Convergence Analysis	21
2.5	Studies on Boundary Conditions and Friction	25
2.5.1	Part A: Reflecting Wall Boundary Conditions	25
2.5.2	Part B: Linear Friction	26
2.5.3	Numerical Results and Physical Interpretation	27
3	Homework 3	32
3.1	Finite Volume Approximation	32
3.1.1	The Mathematical Model	32
3.1.2	FVM Discretization	33
3.1.3	First-Order Godunov Scheme	33
3.1.4	Second-Order Lax-Wendroff Reconstruction	34
3.2	Riemann Problems and Physical Interpretation	35
3.2.1	Dam Break and Reverse Dam Break	35
3.2.2	Stationary Shock (Hydraulic Jump)	36
3.3	Modified Pressure Flux	38
3.3.1	Analytical Formulation	38
3.3.2	Numerical Results and Discussion	39

1

HOMEWORK 1

1.1 WEAK FORM AND FE DISCRETIZATION

In this chapter, the weak formulation and the corresponding finite element (FE) discretization of the boundary value problem introduced in Homework 1 are derived.

1.1.1 STRONG FORMULATION

The one-dimensional wave propagation problem is considered:

$$\begin{cases} \rho(x)\omega^2 u(x) + \frac{\partial}{\partial x} \left(\mu(x) \frac{\partial u(x)}{\partial x} \right) = f & x \in (0, L) \\ \mu(0) \frac{\partial u(x)}{\partial x} \Big|_{x=0} = g_N \\ u(L) - \sqrt{\frac{\mu(L)}{\rho(L)}} \frac{i}{\omega} \frac{\partial u(x)}{\partial x} \Big|_{x=L} = g_A \end{cases} \quad (1.1)$$

In this report, the following notational simplifications are adopted:

- $u = u(x)$ and, analogously, for all functions depending on time;
- $u' = u'(x) = \frac{\partial u(x)}{\partial x}$ and, analogously, for all spatial derivatives;
- $u'(0) = \frac{\partial u(x)}{\partial x} \Big|_{x=0}$ and, analogously, for derivatives calculated in $x = L$.

The second boundary condition is an impedance (or absorbing) condition. It is useful to rewrite it in order to obtain $u'(L)$ explicitly. From the relation above, it follows that:

$$u'(L) = -i\omega \sqrt{\frac{\rho(L)}{\mu(L)}} (u(L) - g_A)$$

For convenience, the impedance coefficient is introduced as

$$\beta = \sqrt{\rho(L)\mu(L)}$$

1.1.2 WEAK FORMULATION

To obtain the weak formulation, the strong form of the equation is multiplied by a test function $v(x) \in H^1(0, L)$ and integrated over $(0, L)$:

$$\int_0^L \rho\omega^2 u v dx + \int_0^L \frac{d}{dx} (\mu u') v dx = \int_0^L f v dx$$

The second term is then integrated by parts:

$$\int_0^L \frac{d}{dx}(\mu u') v dx = [\mu u' v]_0^L - \int_0^L \mu u' v' dx$$

Substituting into the previous expression yields:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx + [\mu u' v]_0^L = \int_0^L f v dx$$

The boundary contributions are incorporated using the boundary conditions. At $x = 0$, it holds that:

$$\mu(0)u'(0)v(0) = g_N v(0) \implies -\mu(0)u'(0)v(0) = -g_N v(0)$$

At $x = L$, using $u'(L)$ from the impedance condition, one obtains:

$$\mu(L)u'(L)v(L) = -i\omega\beta u(L)v(L) + i\omega\beta g_A v(L)$$

Collecting all terms, the weak formulation becomes:

$$\omega^2 \int_0^L \rho u v dx - \int_0^L \mu u' v' dx - i\omega\beta u(L)v(L) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L) \quad (1.2)$$

It is convenient to identify the mass term, stiffness term and the contributions arising from the boundary conditions. The bilinear forms are defined as:

$$\mathcal{M}(u, v) = \int_0^L \rho u v dx, \quad \mathcal{A}(u, v) = \int_0^L \mu u' v' dx + i\omega\beta u(L)v(L)$$

and the linear functional as:

$$\mathcal{F}(v) = \int_0^L f v dx + g_N v(0) - i\omega\beta g_A v(L)$$

Thus, the weak formulation can be written compactly as:

$$\omega^2 \mathcal{M}(u, v) - \mathcal{A}(u, v) = \mathcal{F}(v) \quad \forall v \in H^1(0, L) \quad (1.3)$$

1.1.3 FINITE ELEMENT DISCRETIZATION AND ALGEBRAIC SYSTEM

A partition of the interval $(0, L)$ into N elements is introduced and the finite dimensional space $V_h \subset H^1(0, L)$, spanned by continuous, piecewise linear basis functions $\{\varphi_j\}_{j=1}^N$, is considered. The solution and the test function are approximated as:

$$u_h = u_h(x) = \sum_{j=1}^N u_j(x) \varphi_j(x), \quad v_h = v_h(x) = \varphi_i(x)$$

Substituting u_h and v_h into the equation (1.2) yields the discrete system. The entries of the mass matrix, stiffness matrix and load vector are then defined.

- Mass matrix:

$$M_{ij} = \int_0^L \varphi_i \varphi_j dx$$

- Stiffness matrix and impedance contribution:

$$A_{ij} = \int_0^L \mu \varphi_i' \varphi_j' dx + i\omega\beta \varphi_i(L) \varphi_j(L)$$

Note that $\varphi_i(L) \neq 0$ is only for the basis function associated with the last node of the mesh. Therefore, the impedance term modifies only the entry A_{NN} .

- Load vector:

$$F_i = \int_0^L f \varphi_i dx + g_N \varphi_i(0) - i\omega\beta g_A \varphi_i(L)$$

Here $\varphi_i(0)$ is nonzero only for the basis function associated with the first node, while $\varphi_i(L)$ is nonzero only for the last one.

Collecting all contributions, the linear system is obtained:

$$\omega^2 \mathbf{M} \underline{u} - \mathbf{A} \underline{u} = \underline{F} \quad (1.4)$$

which corresponds to the discretized weak formulation and matches the form requested in the homework assignment. This completes the derivation of the weak formulation and its finite element discretization using linear basis functions.

1.2 MATLAB SOLVER

This section describes the MATLAB implementation used to solve the algebraic system derived in section 1.1 and introduces a verification procedure to assess the correctness of the finite element (FE) code. In particular, the Method of Manufactured Solutions (MMS) is adopted, which consists of prescribing an exact solution and deriving the corresponding forcing terms and boundary data. This provides a controlled test case where the numerical approximation can be directly compared against a known reference.

1.2.1 VERIFICATION TEST SETUP

Considering the one-dimensional domain and constant physical parameters

$$\Omega = (0, 1), \quad L = 1, \quad \rho = 1, \quad \mu = 1$$

For the verification test, the manufactured exact solution is imposed

$$u_{\text{ex}}(x) = \sin(\omega x), \quad \omega = 5\pi$$

Internal forcing term Starting from the strong form in equation (1.1), the governing equation can be written as

$$\rho \omega^2 u(x) + \frac{d}{dx} (\mu u'(x)) = f(x) \quad (1.5)$$

where $u' = \frac{du}{dx}$. Computing the derivatives of the manufactured solution yields.

$$u'_{\text{ex}}(x) = \omega \cos(\omega x), \quad u''_{\text{ex}}(x) = -\omega^2 \sin(\omega x)$$

Since μ is constant, $\frac{d}{dx} (\mu u') = \mu u''$. Substituting u_{ex} into equation (1.5) with $\rho = \mu = 1$ yields

$$f(x) = \omega^2 \sin(\omega x) + u''_{\text{ex}}(x) = \omega^2 \sin(\omega x) - \omega^2 \sin(\omega x) = 0$$

Therefore, the internal source term vanishes:

$$f(x) = 0 \quad (1.6)$$

Neumann condition The Neumann boundary condition in equation (1.1) at $x = 0$ reads

$$\mu(0)u'(0) = g_N \quad (1.7)$$

Using $u'_{\text{ex}}(x) = \omega \cos(\omega x)$ yields

$$u'_{\text{ex}}(0) = \omega \cos(0) = \omega, \quad \mu(0) = 1$$

hence

$$g_N = \omega = 5\pi \quad (1.8)$$

Impedance condition The impedance boundary condition in equation (1.1) at $x = L$ is

$$u(L) - \sqrt{\frac{\mu(L)}{\rho(L)}} \frac{i}{\omega} u'(L) = g_A \quad (1.9)$$

Since $\rho(L) = \mu(L) = 1$, the impedance coefficient simplifies to

$$\sqrt{\frac{\mu(L)}{\rho(L)}} = 1$$

Evaluating the manufactured solution at $L = 1$ gives

$$u_{\text{ex}}(1) = \sin(5\pi) = 0, \quad u'_{\text{ex}}(1) = 5\pi \cos(5\pi) = -5\pi$$

Substituting into the boundary condition yields

$$g_A = u_{\text{ex}}(1) - \frac{i}{\omega} u'_{\text{ex}}(1) = 0 - \frac{i}{5\pi} (-5\pi) = i$$

Therefore, the impedance boundary datum is

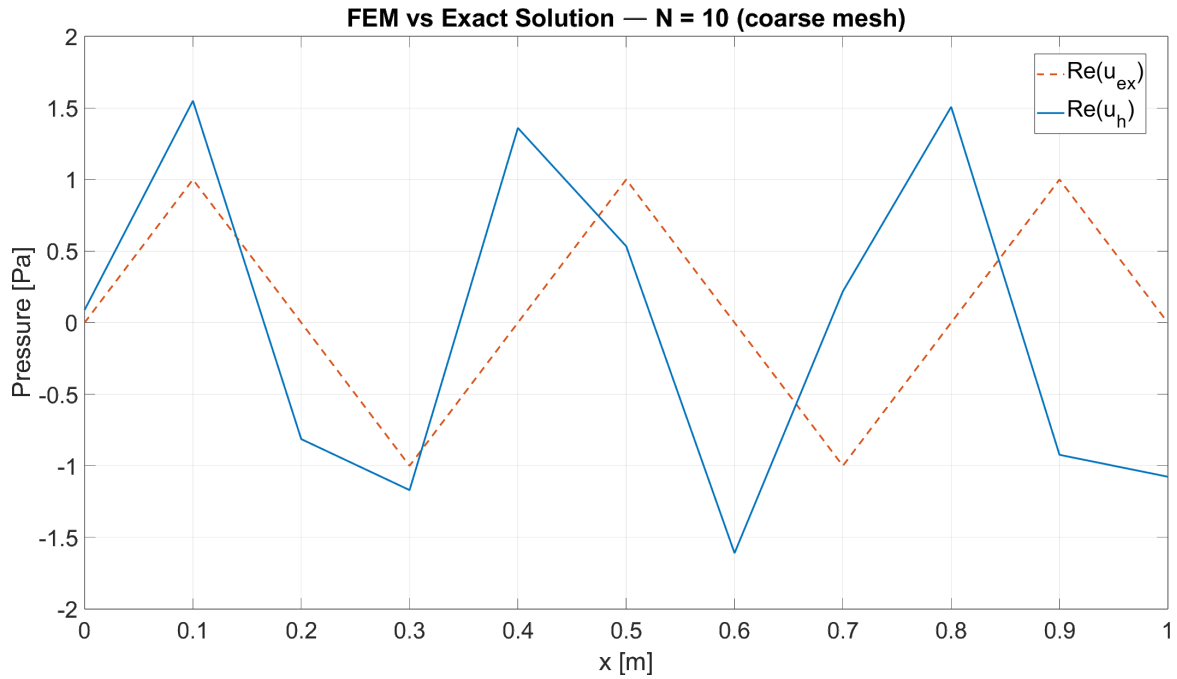
$$g_A = i \quad (1.10)$$

1.2.2 NUMERICAL RESULTS

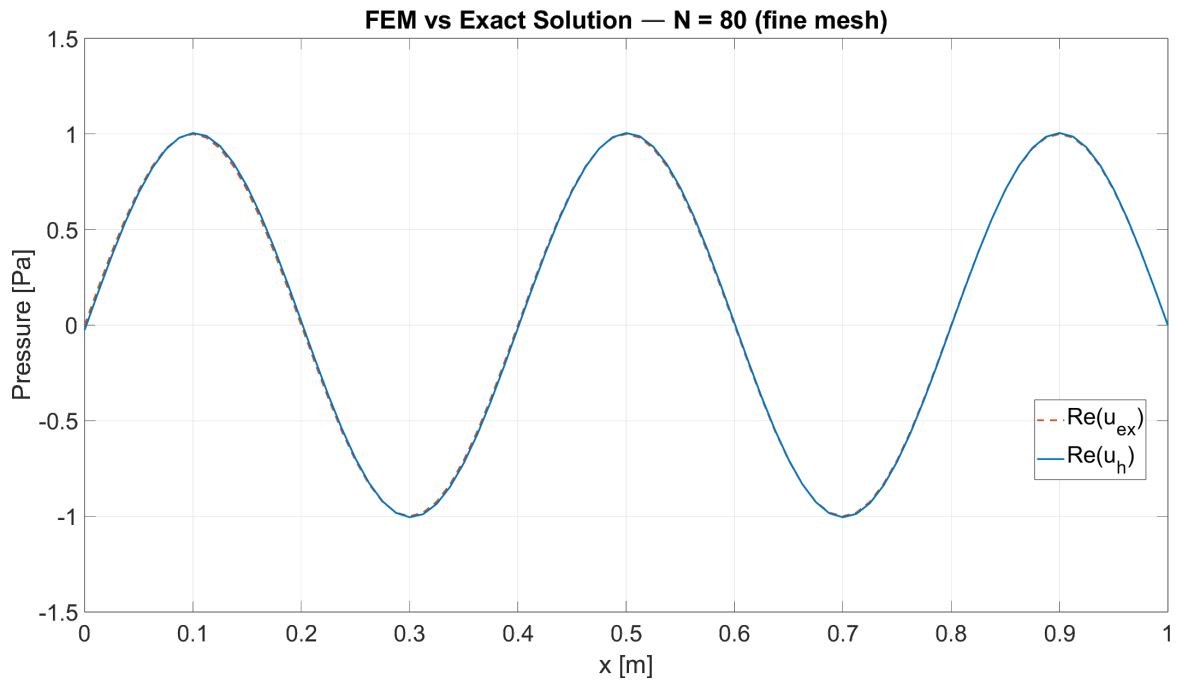
The algebraic system (1.4) is assembled and solved in MATLAB for a sequence of uniform meshes with $N \in \{10, 20, 40, 80\}$ elements. Since $\omega = 5\pi$, the wavelength of the exact solution is

$$\lambda = \frac{2\pi}{\omega} = \frac{2}{5} = 0.4 \text{ m}$$

so the domain $(0, 1)$ contains exactly 2.5 full wavelengths. On the coarsest mesh ($N = 10, h = 0.1$), each wavelength is covered by only $\lambda/h = 4$ elements, which is insufficient to resolve the oscillatory behavior of u_{ex} . On the finest mesh ($N = 80, h = 0.0125$), the resolution rises to 32 elements per wavelength, yielding an accurate approximation. Figure 1.1 compares the real part of the FE solution u_h with the exact solution u_{ex} on the coarse and fine meshes respectively. As expected, the coarse-mesh solution captures the correct frequency but with a noticeable phase and amplitude error, while the fine-mesh solution overlaps the exact curve almost perfectly.



(a) On the coarse mesh ($N = 10$, $h = 0.1$). The four elements per wavelength are insufficient to resolve the sinusoidal shape, resulting in a visible phase shift and amplitude overshoot



(b) On the fine mesh ($N = 80$, $h = 0.0125$). With 32 elements per wavelength the numerical and exact curves are virtually indistinguishable

Figure 1.1: Real part of the FE pressure field compared with the exact solution $u_{\text{ex}} = \sin(\omega x)$

1.2.3 CONVERGENCE ANALYSIS

To quantify the approximation quality, the $L^2(0, 1)$ error is computed for each mesh via numerical quadrature:

$$\|u - u_h\|_{L^2(0,L)} = \left(\int_0^L |u_{\text{ex}}(x) - u_h(x)|^2 dx \right)^{1/2}$$

The theoretical a-priori estimate for continuous piecewise-linear ($\mathbb{P}_1$) Galerkin approximations of elliptic problems guarantees

$$\|u - u_h\|_{L^2(0,L)} \leq Ch^2|u|_{H^2(0,L)} \quad (1.11)$$

so one expects a convergence rate of order 2 in the mesh size h . Table 1.1 reports the computed L^2 errors and the corresponding numerical convergence rates p , estimated from consecutive mesh pairs as

$$p = \frac{\log(e_i/e_{i+1})}{\log(h_i/h_{i+1})}$$

N	h	$\ u - u_h\ _{L^2}$	Rate p
10	0.1000	8.21×10^{-1}	—
20	0.0500	2.05×10^{-1}	2.04
40	0.0250	5.12×10^{-2}	2.21
80	0.0125	1.28×10^{-2}	3.05

Table 1.1: L^2 errors and estimated convergence rates for the MMS verification test with $u_{\text{ex}} = \sin(5\pi x)$ and NI boundary conditions

The numerical rates confirm the expected $\mathcal{O}(h^2)$ convergence, consistent with the theoretical estimate (1.11) and with the use of $\mathbb{P}_1$ basis functions. The log-log convergence plot in figure 1.2 shows that the error decreases along a straight line of slope 2, parallel to the reference $\mathcal{O}(h^2)$ dashed line. This agreement between theory and numerics validates the correctness of the MATLAB implementation.

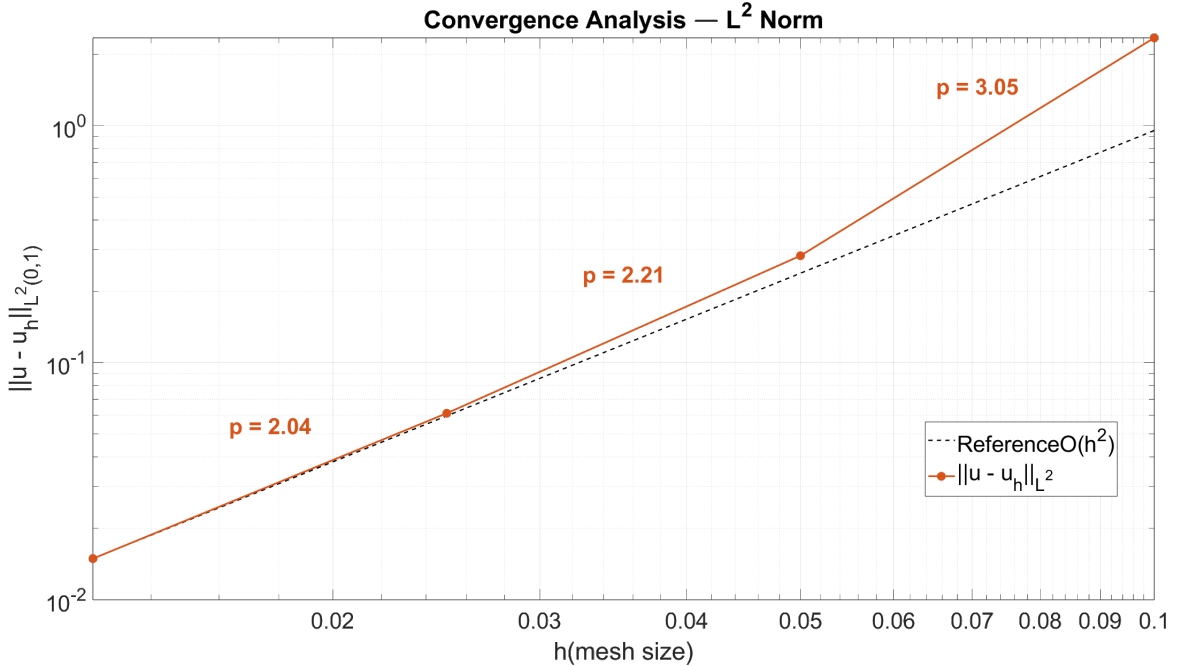


Figure 1.2: Log-log convergence plot of the L^2 error $\|u - u_h\|_{L^2(0,1)}$ as a function of the mesh size h for the MMS test with $u_{\text{ex}} = \sin(5\pi x)$. The dashed line represents the reference slope $\mathcal{O}(h^2)$ and the annotated rates confirm quadratic convergence for $\mathbb{P}_1$ elements

1.3 WAVE REFLECTION AT AN IMPEDANCE JUMP

The propagation of a time-harmonic field in a heterogeneous one-dimensional medium is considered, with a discontinuity in the material properties located at mid-domain. The main effect of such discontinuity is the generation of a partially reflected wave due to an impedance mismatch.

1.3.1 PROBLEM SETUP

The spatial domain is defined as $\Omega = (0, 1)$ with constant density $\rho(x) = 1$. The stiffness is assumed piecewise constant and discontinuous at $x = 0.5$, according to

$$\mu(x) = \begin{cases} 4, & x \leq 0.5 \\ 1, & x > 0.5 \end{cases} \quad (1.12)$$

No volumetric source term is applied, hence $f(x) = 0$. The excitation is imposed at the left boundary through a Neumann condition $\mu(0)u'(0) = g_N$ with $g_N = 1$, whereas at the right boundary a perfectly absorbing condition is prescribed by setting $g_A = 0$. This choice prevents spurious reflections at $x = 1$, so that the observed reflected component is generated only at the internal interface $x = 0.5$.

1.3.2 MECHANICAL IMPEDANCE OF THE TWO MEDIA

For one-dimensional wave propagation, reflection and transmission at an interface are governed by the characteristic (mechanical) impedance of the medium, defined as

$$Z = \sqrt{\rho\mu} \quad (1.13)$$

Since $\rho = 1$ throughout Ω , the impedance depends only on μ . In the left subdomain $x \leq 0.5$ it follows that $Z_1 = \sqrt{1 \cdot 4} = 2$, while in the right subdomain $x > 0.5$ one obtains $Z_2 = \sqrt{1 \cdot 1} = 1$. The corresponding phase velocities are $c_1 = \sqrt{\mu_1/\rho} = 2$ and $c_2 = \sqrt{\mu_2/\rho} = 1$ respectively.

1.3.3 REFLECTION COEFFICIENT AT THE INTERFACE

When a right-going incident wave traveling in medium 1 reaches the interface with medium 2, continuity of displacement u and traction $\mu u'$ implies the formation of both a reflected and a transmitted component. The reflection coefficient for incidence from medium 1 to medium 2 is given by

$$R = \frac{Z_1 - Z_2}{Z_1 + Z_2} \quad (1.14)$$

Substituting $Z_1 = 2$ and $Z_2 = 1$ yields

$$R = \frac{2 - 1}{2 + 1} = \frac{1}{3}$$

Therefore, a partial reflection is expected at $x = 0.5$ with positive sign, i.e., without phase inversion.

1.3.4 EXPECTED QUALITATIVE BEHAVIOR IN AMPLITUDE AND PHASE

Distinct qualitative behaviors are expected on the two sides of the interface. In the left region $x < 0.5$, the superposition of the incident and reflected waves produces an interference pattern, corresponding to a partially standing-wave regime. As a consequence, oscillations are expected in both the amplitude profile $|u(x)|$ and the phase $\arg(u(x))$, which will be irregular and non-monotone. In the right region $x > 0.5$, the transmitted wave propagates toward $x = 1$. Due to the perfectly absorbing condition $g_A = 0$, no reflection is generated at the right boundary and the solution resembles a single traveling wave. Hence, the amplitude is expected to be comparatively smooth and the phase to grow linearly and monotonically with slope $\omega/c_2 = \omega$. At the interface $x = 0.5$, a discontinuity in the spatial derivative of the phase is expected, reflecting the change in phase velocity from $c_1 = 2$ to $c_2 = 1$: since the phase slope of a traveling wave is ω/c , the slope in the right subdomain is twice that in the left subdomain for a purely traveling wave.

1.3.5 NUMERICAL RESULTS

The problem is solved numerically using the MATLAB solver described in chapter 1.1, with $N = 200$ elements to ensure sufficient resolution across both subdomains. The stiffness coefficient is implemented as a piecewise function evaluated at the quadrature nodes of each element. Figure 1.3 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ of the computed solution.

1.3.6 PHYSICAL INTERPRETATION

The numerical results are in full agreement with the analytical predictions derived in the previous sections. Three distinct regions can be identified in the solution.

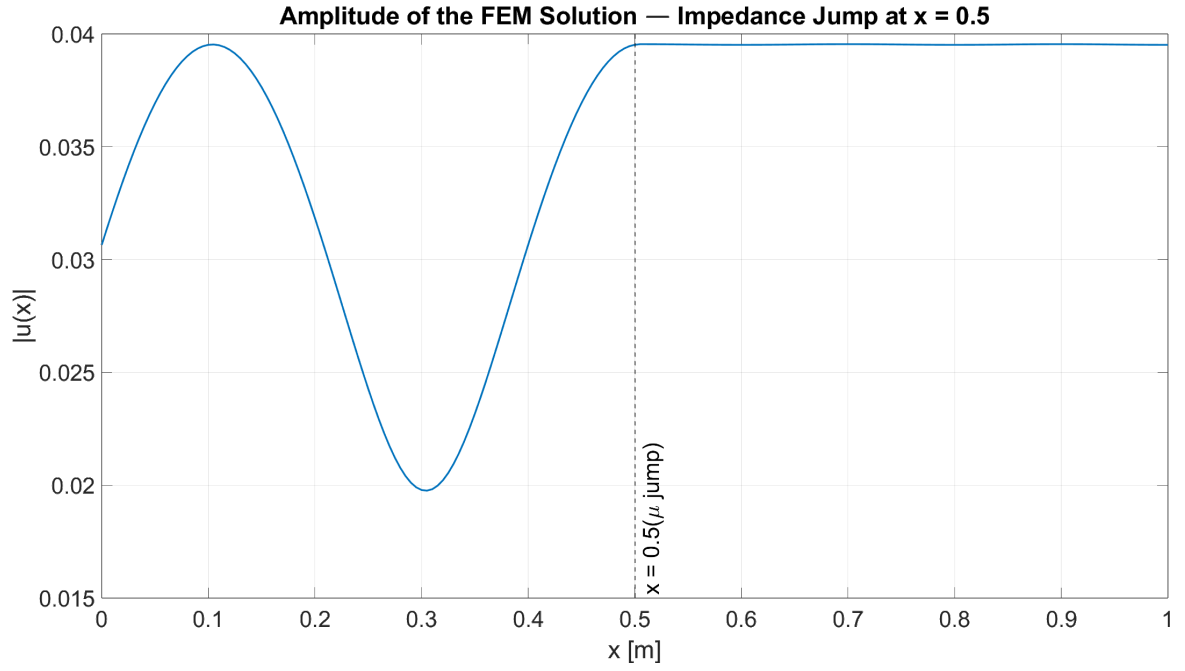
Left subdomain $x < 0.5$ — partial standing wave The amplitude profile $|u(x)|$ exhibits oscillations with alternating local maxima and minima, which are the signature of the interference between the rightward-traveling incident wave and the leftward-traveling reflected wave generated at the interface. The phase is irregular and non-monotone, consistently with the superposition of two counter-propagating components. The reflection coefficient $R = 1/3$ implies that approximately 11% of the incident energy is reflected back into the left subdomain, producing a moderately pronounced standing-wave pattern.

Right subdomain $x > 0.5$ — traveling wave Beyond the interface, the amplitude is smooth and the phase grows linearly with slope $\omega/c_2 = \omega$, confirming the presence of a single right-going traveling wave. The absorbing boundary condition at $x = 1$ effectively prevents any reflection from the right end, so no additional interference pattern appears in this region.

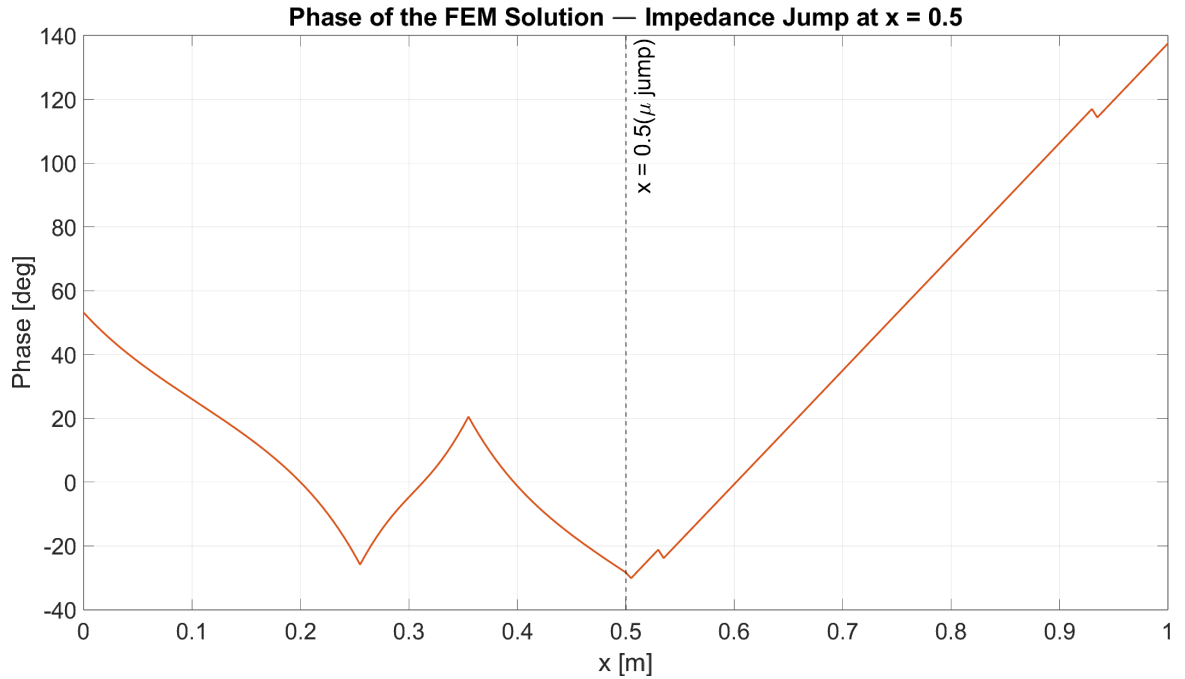
Interface $x = 0.5$ — phase velocity change A clear kink in the phase is visible at the interface. Since the phase slope of a traveling wave equals ω/c , the ratio of the slopes on the two sides of the interface is

$$\frac{(\omega/c_2)}{(\omega/c_1)} = \frac{c_1}{c_2} = \frac{2}{1} = 2$$

confirming that the phase advances twice as fast per unit length in the right medium, consistently with $c_2 = 1 < c_1 = 2$.



(a) Amplitude $|u(x)|$ of the numerical solution. The oscillatory pattern in the left subdomain ($x < 0.5$) is the signature of the standing-wave component generated by the partial reflection at the interface, while the smoother decay in the right subdomain ($x > 0.5$) corresponds to the purely transmitted wave



(b) Unwrapped phase $\arg(u(x))$ of the numerical solution. The irregular, non-monotone behavior for $x < 0.5$ is due to the interference between the incident and reflected waves, while the linear monotone growth for $x > 0.5$ is consistent with a single right-going traveling wave. The kink at $x = 0.5$ reflects the change in phase velocity between the two media

Figure 1.3: Amplitude and unwrapped phase of the FE solution for the impedance-jump problem with μ as in equation (1.12), $\rho = 1$, $g_N = 1$, $g_A = 0$ and $\omega = 5\pi$

1.4 NUMERICAL DISPERSION ANALYSIS

It is well known that spatial discretization introduces a dispersion error: the discrete solution does not propagate with the same phase velocity as the continuous (physical) wave. In

the frequency domain, this phenomenon is conveniently quantified by comparing the continuous wavenumber k with the discrete (numerical) wavenumber k_h , associated with the finite-dimensional approximation. The resulting discrepancy represents the numerical dispersion of the method.

1.4.1 EIGENVALUE PROBLEM SETTING

To isolate dispersion effects from external excitations, the homogeneous problem is considered by setting $f = 0$ and imposing homogeneous boundary data. In particular, a homogeneous Neumann condition is prescribed at the left boundary, namely $g_N = 0$, which implies $\mu u'(0) = 0$, while a homogeneous Dirichlet condition is imposed at the right boundary, $u(L) = 0$. Under these assumptions, the Helmholtz model reduces to an eigenvalue problem for the spatial operator.

1.4.2 CONTINUOUS AND DISCRETE FORMULATIONS

In the continuous setting and for constant coefficients $\rho = \mu = 1$, the eigenmodes satisfy the classical second-order equation

$$-u''(x) = k^2 u(x) \quad (1.15)$$

where k^2 denotes the exact eigenvalues of the continuous problem and k is the corresponding continuous wavenumber. By applying the finite element discretization derived in section 1.1 and restricting the solution to the discrete space, the homogeneous algebraic system assumes the form of a generalized eigenvalue problem. Denoting by A the stiffness matrix and by M the mass matrix, the discrete modes satisfy

$$A\mathbf{u} = k_h^2 M\mathbf{u} \quad (1.16)$$

where k_h^2 are the discrete eigenvalues and k_h is the associated numerical wavenumber. The dispersion error is therefore interpreted as the difference between the spectra of the continuous and discrete operators, expressed in terms of wavenumbers.

1.4.3 DISPERSION CURVE AND NUMERICAL RESULTS

The accuracy of the method is assessed by introducing the dimensionless resolution parameter h/λ , where h denotes the characteristic mesh size and λ the wavelength. The numerical dispersion is represented through the dispersion curve, defined as the ratio k_h/k plotted as a function of h/λ . The generalized eigenvalue problem (1.16) is solved in MATLAB for a sequence of meshes with $N \in \{10, 20, 40, 80, 160\}$ elements. For each discrete eigenvalue k_h^2 , the corresponding exact wavenumber k is identified analytically and the ratio k_h/k is computed. Only modes with $h/\lambda < 0.5$ are retained to exclude poorly resolved optical branches; in practice, the data extend up to $h/\lambda \approx 0.36$ due to the limited number of available eigenvalues on the coarser meshes. The resulting dispersion curve is shown in figure 1.4.

1.4.4 PHYSICAL INTERPRETATION

The dispersion curve in figure 1.4 exhibits a monotonically decreasing trend, in full agreement with the theoretical prediction for $\mathbb{P}_1$ Galerkin approximations and three distinct resolution regimes can be identified.

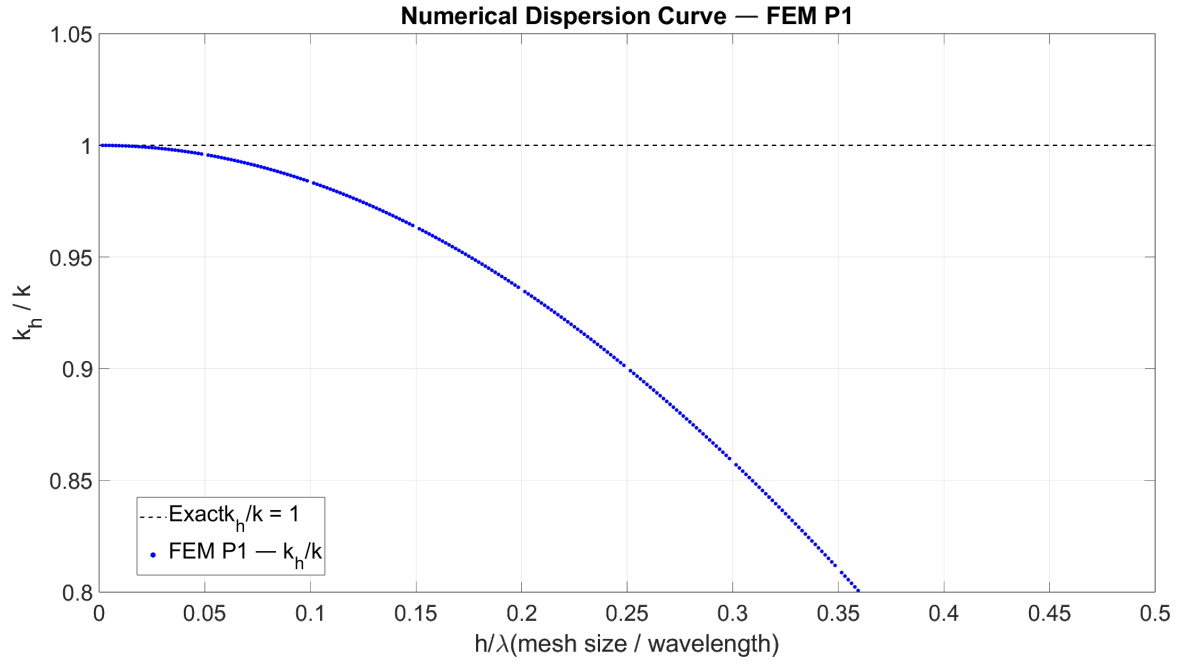


Figure 1.4: Numerical dispersion curve k_h/k as a function of the resolution parameter h/λ for the $\mathbb{P}_1$ FEM. The dashed line at $k_h/k = 1$ represents the exact (non-dispersive) behavior. As $h/\lambda \rightarrow 0$ the FEM solution converges to the exact wavenumber, while coarser meshes produce an increasingly large underestimation of k

Fine meshes — $h/\lambda \rightarrow 0$ As the mesh is refined, $k_h/k \rightarrow 1$ and the FEM reproduces the continuous wavenumber with negligible error. Below $h/\lambda \approx 0.05$, corresponding to roughly 20 or more elements per wavelength, the dispersion error is practically zero and the numerical phase velocity coincides with the physical one.

Intermediate resolution For $h/\lambda = 0.1$, i.e., approximately 10 elements per wavelength, the ratio is $k_h/k \approx 0.903$, corresponding to a dispersion error of about 1.7%. This level of resolution is commonly considered a practical minimum for engineering applications where moderate accuracy is acceptable.

Coarse meshes — large h/λ At $h/\lambda = 0.25$, corresponding to only 4 elements per wavelength, the ratio drops to $k_h/k \approx 0.90$, meaning a 10% underestimation of the physical wavenumber. In this regime, the FEM underestimates k_h and consequently overestimates the phase velocity, a well-known artifact of low-order Galerkin discretizations. This explains qualitatively the phase shift and amplitude distortion already observed on the coarse mesh in section 1.1.

General rule of thumb The analysis confirms that at least 10 elements per wavelength ($h/\lambda \leq 0.1$) are required to keep the dispersion error below 2% and that 20 or more elements per wavelength ($h/\lambda \leq 0.05$) are needed for high-fidelity simulations. These thresholds are consistent with the standard engineering guidelines for $\mathbb{P}_1$ FEM applied to wave propagation problems.

1.5 PERFECTLY MATCHED LAYER (PML)

In order to reduce spurious reflections at the right boundary, an alternative to the absorbing (impedance) boundary condition at $x = 1$ is considered. The computational domain is extended by adding an absorbing layer, referred to as a Perfectly Matched Layer (PML), in the region $x \in (1, 1 + L_{\text{PML}})$. The PML is designed to damp outgoing waves without reflecting them back into the physical domain $\Omega = (0, 1)$.

1.5.1 DAMPING PROFILE

The attenuation mechanism is introduced through a spatially varying damping function $\sigma(x)$. In order to achieve effective absorption while minimizing numerical reflections due to abrupt coefficient variations, $\sigma(x)$ is required to increase smoothly from zero at the interface $x = 1$. A parabolic profile is adopted, defined as

$$\sigma(x) = \begin{cases} 0 & \text{for } x \leq 1 \\ \sigma_{\max} \left(\frac{x-1}{L_{\text{PML}}} \right)^2 & \text{for } x > 1 \end{cases} \quad (1.17)$$

1.5.2 COMPLEX STRETCHING AND MODIFIED COEFFICIENTS

The PML formulation is obtained by introducing the complex stretching function

$$s(x) = 1 - i \frac{\sigma(x)}{\omega} \quad (1.18)$$

Within the layer, the Helmholtz model is recast by replacing the physical parameters ρ and μ with complex-valued equivalents,

$$\tilde{\rho}(x) = \rho(x)s(x), \quad \tilde{\mu}(x) = \frac{\mu(x)}{s(x)} \quad (1.19)$$

This choice preserves the structure of the governing equation while introducing attenuation through complex coefficients.

1.5.3 IMPEDANCE MATCHING AND EXPONENTIAL DECAY

A key property of the PML is that it is perfectly matched at the interface $x = 1$. In particular, the characteristic impedance in the PML region satisfies

$$Z_{\text{PML}} = \sqrt{\tilde{\rho}(x)\tilde{\mu}(x)} = \sqrt{\rho(x)s(x)\frac{\mu(x)}{s(x)}} = \sqrt{\rho(x)\mu(x)} = Z \quad (1.20)$$

which shows that the impedance is continuous across the interface. This condition implies, at the continuous level, perfect transmission and therefore the absence of reflected waves at $x = 1$. On the other hand, the local wavenumber becomes complex. Defining $k = \omega\sqrt{\rho/\mu}$ as the physical wavenumber, the PML wavenumber is obtained as

$$k_{\text{PML}}(x) = \omega\sqrt{\frac{\tilde{\rho}(x)}{\tilde{\mu}(x)}} = \omega\sqrt{\frac{\rho(x)s(x)}{\mu(x)/s(x)}} = \omega\sqrt{\frac{\rho(x)}{\mu(x)}}s(x) = ks(x) \quad (1.21)$$

Since $s(x) = 1 - i\sigma(x)/\omega$ has a negative imaginary part for $\sigma(x) \geq 0$, the transmitted wave in the layer experiences an exponential decay in amplitude as x increases, leading to strong attenuation before reaching the truncated boundary.

1.5.4 BOUNDARY CONDITIONS ON THE EXTENDED DOMAIN

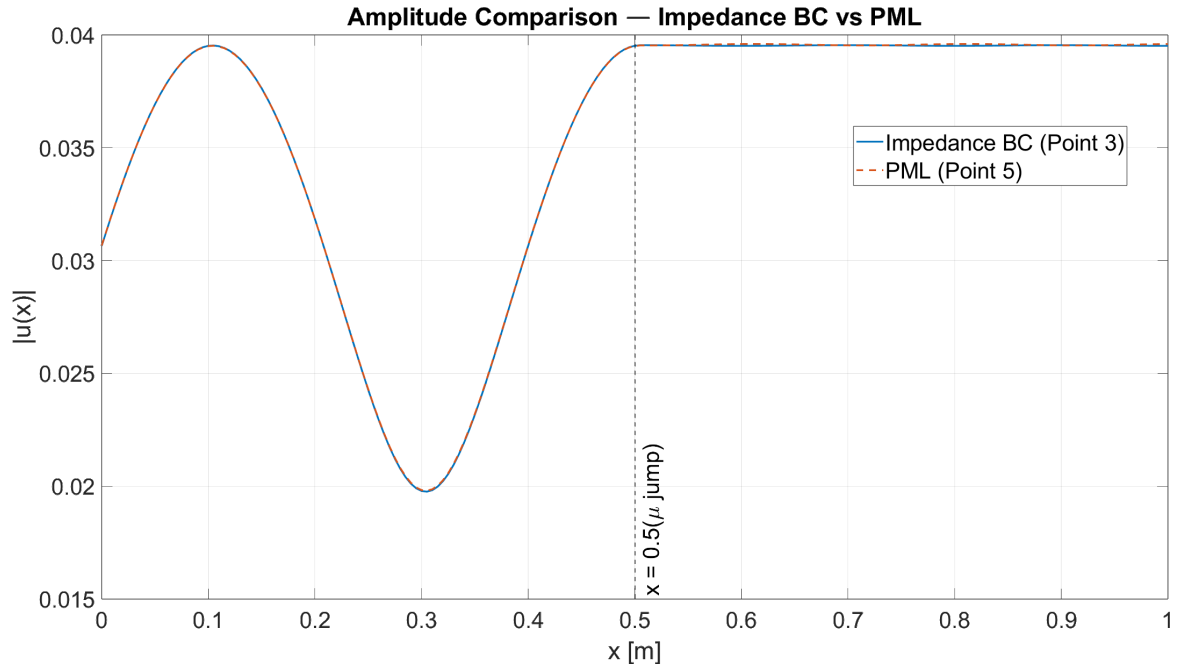
The excitation at the left boundary is kept unchanged and the Neumann condition is imposed at $x = 0$. The right boundary is moved to the end of the PML, located at $x = 1 + L_{\text{PML}}$, where a homogeneous Dirichlet condition is prescribed, $u(1 + L_{\text{PML}}) = 0$. This truncation is justified by the assumption that the outgoing wave has been sufficiently damped by the PML and is therefore negligible at the outer boundary.

1.5.5 NUMERICAL RESULTS AND COMPARISON

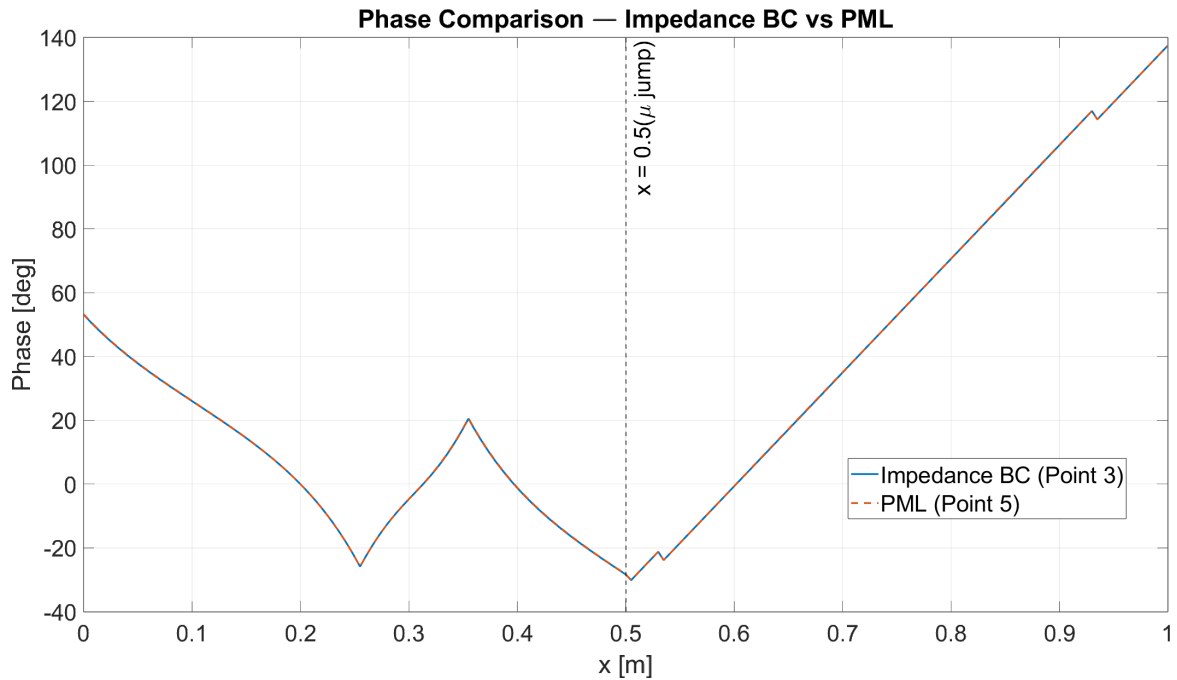
The PML approach is validated by comparing its solution against the impedance boundary condition result obtained in chapter 1.3, for the same physical setup: $\mu(x)$ as in equation (1.12), $\rho = 1$, $g_N = 1$, $\omega = 5\pi$ and $N = 200$ elements in the physical domain $(0, 1)$. The PML layer extends over $x \in (1, 1.2)$ with $L_{\text{PML}} = 0.2$ and $\sigma_{\text{max}} = 100$, discretized with an additional 40 elements, for a total of 240 degrees of freedom on the extended domain. Figure 1.5 reports the amplitude $|u(x)|$ and the unwrapped phase $\arg(u(x))$ restricted to the physical domain $(0, 1)$, comparing the two methods.

1.5.6 PHYSICAL INTERPRETATION

The near-perfect agreement between the two solutions confirms that both methods correctly model the same physical phenomenon, a perfectly absorbing right boundary, through two fundamentally different strategies. The impedance boundary condition enforces the non-reflecting constraint analytically at a single point $x = 1$, while the PML distributes the attenuation numerically over an extended layer. The quasi-perfect overlap in both amplitude and phase throughout the physical domain validates the correctness of both implementations. The only visible discrepancy appears in the amplitude for $x > 0.5$, where the PML curve is marginally higher than the impedance BC result. This is a well-known discrete artifact: despite the impedance-matching property (1.20) holding exactly at the continuous level, the discretization introduces a small spurious reflection at the interface $x = 1$ whose magnitude depends on the choice of σ_{max} and L_{PML} . Increasing either parameter reduces the residual reflection, at the cost of a larger computational domain.



(a) Amplitude $|u(x)|$. Both curves overlap throughout $(0, 1)$: the oscillatory interference pattern for $x < 0.5$ and the smooth transmitted field for $x > 0.5$ are identical in the two approaches. A slight overestimation of the PML curve (dashed) is visible for $x > 0.5$, attributable to a small discrete numerical reflection at the PML interface



(b) Unwrapped phase $\arg(u(x))$. The irregular non-monotone behavior for $x < 0.5$ and the linear growth for $x > 0.5$ are virtually indistinguishable between the impedance BC and the PML solutions

Figure 1.5: Comparison of the FE solution obtained with the impedance boundary condition (Point 3, section 1.3) and the PML formulation (Point 5, section 1.5), restricted to the physical domain $\Omega = (0, 1)$

2

HOMEWORK 2

2.1 EQUIVALENCE TO THE SECOND-ORDER WAVE EQUATION

The linearized shallow water equations govern the propagation of surface waves in a shallow domain $(x, t) \in (0, L) \times (0, T)$. The equivalence between the first-order coupled system and the classical second-order wave equation is established by eliminating one of the two unknowns.

2.1.1 STRONG FORMULATION

The governing system reads:

$$\begin{cases} \eta_t(x, t) + q_x(x, t) = 0 \\ q_t(x, t) + gH\eta_x(x, t) = 0 \end{cases} \quad (2.1)$$

where $\eta(x, t)$ is the free surface elevation, $q(x, t)$ is the discharge, H is the mean water depth, and g is the gravitational acceleration. The system is supplemented by periodic boundary conditions:

$$\eta(0, t) = \eta(L, t), \quad q(0, t) = q(L, t) \quad (2.2)$$

2.1.2 DECOUPLING THE SYSTEM

The discharge variable $q(x, t)$ is eliminated by cross-differentiation. Differentiating the first equation of (2.1) with respect to time t :

$$\eta_{tt}(x, t) + q_{xt}(x, t) = 0 \implies \eta_{tt}(x, t) = -q_{xt}(x, t) \quad (2.3)$$

Differentiating the second equation of (2.1) with respect to space x :

$$q_{tx}(x, t) + gH\eta_{xx}(x, t) = 0 \implies q_{tx}(x, t) = -gH\eta_{xx}(x, t) \quad (2.4)$$

Assuming sufficient regularity, specifically $q \in C^2((0, L) \times (0, T))$, Schwarz's theorem yields $q_{xt} = q_{tx}$. Substituting (2.4) into (2.3) results in the standard second-order wave equation:

$$\eta_{tt}(x, t) = gH\eta_{xx}(x, t) \quad \text{for } (x, t) \in (0, L) \times (0, T) \quad (2.5)$$

2.1.3 DETERMINATION OF SUITABLE BOUNDARY CONDITIONS

The periodic boundary conditions on the coupled variables (η, q) must be translated into conditions solely on η and its derivatives. The first condition is already explicit:

$$\eta(0, t) = \eta(L, t) \quad (2.6)$$

To convert the second condition $q(0, t) = q(L, t)$, both sides are differentiated with respect to t , yielding $q_t(0, t) = q_t(L, t)$. From the second equation of (2.1), $q_t = -gH\eta_x$. Evaluating at both boundaries and substituting:

$$-gH\eta_x(0, t) = -gH\eta_x(L, t) \implies \eta_x(0, t) = \eta_x(L, t) \quad (2.7)$$

2.1.4 EQUIVALENCE

The first-order system (2.1) is therefore fully equivalent to the second-order acoustic wave equation:

$$\eta_{tt}(x, t) = gH\eta_{xx}(x, t), \quad (x, t) \in (0, L) \times (0, T) \quad (2.8)$$

subject to the periodic boundary conditions $\eta(0, t) = \eta(L, t)$ and $\eta_x(0, t) = \eta_x(L, t)$, derived in (2.6) and (2.7) respectively.

2.2 CONSERVATION OF ENERGY

The conservation of mechanical energy under periodic boundary conditions is established by showing that the time derivative of the prescribed energy functional vanishes identically.

2.2.1 DEFINITION OF THE ENERGY FUNCTIONAL

The energy functional of the shallow water system over the spatial domain $(0, L)$ is defined as:

$$E(t) = \int_0^L \left(\frac{1}{2}g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \quad (2.9)$$

where the first term represents the potential energy associated with the free surface elevation η , and the second term represents the kinetic energy associated with the discharge q . To prove conservation, it suffices to show that $\frac{dE}{dt} = 0$.

2.2.2 TIME DERIVATIVE OF THE ENERGY

Assuming sufficient smoothness of $\eta(x, t)$ and $q(x, t)$, the Leibniz rule permits the exchange of differentiation and integration:

$$\frac{dE}{dt} = \frac{d}{dt} \int_0^L \left(\frac{1}{2}g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx = \int_0^L \frac{\partial}{\partial t} \left(\frac{1}{2}g\eta^2 + \frac{1}{2}\frac{q^2}{H} \right) dx \quad (2.10)$$

Applying the chain rule to compute the partial time derivatives of the integrand yields:

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H}qq_t \right) dx \quad (2.11)$$

2.2.3 SUBSTITUTION OF THE STRONG FORMULATION

Recalling the linearized shallow water equations (2.1):

$$\begin{cases} \eta_t = -q_x \\ q_t = -gH\eta_x \end{cases} \quad (2.12)$$

Substituting these expressions for η_t and q_t into (2.11):

$$\frac{dE}{dt} = \int_0^L \left(g\eta(-q_x) + \frac{1}{H}q(-gH\eta_x) \right) dx \quad (2.13)$$

2.2.4 ALGEBRAIC SIMPLIFICATION AND THE PRODUCT RULE

Factoring out the common constant $-g$, the integral simplifies to:

$$\frac{dE}{dt} = -g \int_0^L (\eta q_x + q \eta_x) dx \quad (2.14)$$

The integrand is recognized as the exact spatial derivative of the product ηq via the product rule $\frac{\partial}{\partial x}(\eta q) = \eta_x q + \eta q_x$. Therefore:

$$\frac{dE}{dt} = -g \int_0^L \frac{\partial}{\partial x}(\eta q) dx \quad (2.15)$$

2.2.5 APPLICATION OF THE FUNDAMENTAL THEOREM OF CALCULUS

Integrating the exact derivative over $[0, L]$ yields the evaluation of the boundary terms:

$$\frac{dE}{dt} = -g \left[\eta(x, t)q(x, t) \right]_{x=0}^{x=L} = -g (\eta(L, t)q(L, t) - \eta(0, t)q(0, t)) \quad (2.16)$$

2.2.6 APPLICATION OF BOUNDARY CONDITIONS

The periodic boundary conditions prescribe $\eta(0, t) = \eta(L, t)$ and $q(0, t) = q(L, t)$. Substituting these into the boundary evaluation:

$$\eta(L, t)q(L, t) - \eta(0, t)q(0, t) = \eta(L, t)q(L, t) - \eta(L, t)q(L, t) = 0 \quad (2.17)$$

2.2.7 ENERGY CONSERVATION

Therefore, the time derivative of the energy is exactly zero:

$$\frac{dE}{dt} = 0 \quad (2.18)$$

This rigorously demonstrates that the energy $E(t)$ defined in (2.9) is conserved for all $t \in (0, T)$ under the prescribed periodic boundary conditions.

2.3 WEAK FORM AND SPECTRAL ELEMENT DISCRETIZATION

A variational formulation of the linearized shallow water system is derived and subsequently discretized using the Spectral Element Method (SEM), leading to a global first-order block ODE system suitable for time integration.

2.3.1 FUNCTION SPACES

The appropriate function space is defined to accommodate the periodic boundary conditions $\eta(0, t) = \eta(L, t)$ and $q(0, t) = q(L, t)$. Solutions are sought in the space of periodic functions with square-integrable first derivatives:

$$V = H_{\text{per}}^1(0, L) = \{v \in H^1(0, L) : v(0) = v(L)\}$$

For any $t \in (0, T)$, it is assumed that $\eta(\cdot, t) \in V$ and $q(\cdot, t) \in V$. Test functions are drawn from the same space: $v, w \in V$.

2.3.2 WEAK FORMULATION

Starting from the strong form of the system:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = 0 \end{cases} \quad (2.19)$$

each equation is multiplied by a test function and integrated over the domain $(0, L)$:

$$\begin{aligned} \int_0^L \eta_t v dx + \int_0^L q_x v dx &= 0 \quad \forall v \in V \\ \int_0^L q_t w dx + gH \int_0^L \eta_x w dx &= 0 \quad \forall w \in V \end{aligned}$$

Boundary Treatment Applying integration by parts to the convective term in the first equation:

$$\int_0^L q_x v dx = [qv]_0^L - \int_0^L q v_x dx$$

Since both q and v belong to V , the boundary term vanishes exactly by the periodicity constraint:

$$[qv]_0^L = q(L)v(L) - q(0)v(0) = 0$$

For the continuous Galerkin SEM with periodic basis functions, it is standard to retain the spatial derivatives on the trial functions. The weak formulation therefore reads: find $\eta, q \in V$ such that for all $v, w \in V$,

$$\begin{cases} \int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \\ \int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \end{cases} \quad (2.20)$$

2.3.3 SPECTRAL ELEMENT DISCRETIZATION

Mesh and Element Partition The domain $\Omega = (0, L)$ is partitioned into N_e non-overlapping elements $\Omega_e = [x_{e-1}, x_e]$. Each element is mapped to the reference element $[-1, 1]$ via an affine transformation.

Basis Functions A finite-dimensional subspace $V_h \subset V$ is introduced. In SEM, the basis functions $\phi_j(x)$ are piecewise continuous polynomials of degree N , specifically Lagrange polynomials constructed over Gauss-Lobatto-Legendre (GLL) nodes. The continuous solutions are approximated as:

$$\eta(x, t) \approx \eta_h(x, t) = \sum_{j=1}^{N_{\text{dof}}} \eta_j(t) \phi_j(x), \quad q(x, t) \approx q_h(x, t) = \sum_{j=1}^{N_{\text{dof}}} q_j(t) \phi_j(x)$$

where N_{dof} is the total number of global degrees of freedom, with the periodic assembly mapping the last node to the first.

2.3.4 GALERKIN SYSTEM AND MATRIX ASSEMBLY

Substituting the discrete approximations and choosing $v = w = \phi_i(x)$, the semi-discrete system becomes:

$$\begin{aligned} \sum_j \dot{\eta}_j \int_0^L \phi_j \phi_i dx + \sum_j q_j \int_0^L \phi'_j \phi_i dx &= 0 \\ \sum_j \dot{q}_j \int_0^L \phi_j \phi_i dx + gH \sum_j \eta_j \int_0^L \phi'_j \phi_i dx &= 0 \end{aligned}$$

The global mass matrix $\mathbf{M}$ and the first-derivative matrix $\mathbf{D}$ are defined by:

$$M_{ij} = \int_0^L \phi_i \phi_j dx, \quad D_{ij} = \int_0^L \phi_i \phi'_j dx$$

Note on GLL Quadrature In SEM, integrals are evaluated using GLL quadrature. The discrete orthogonality property of the Lagrange basis at the GLL nodes renders $\mathbf{M}$ diagonal (lumped), which is advantageous for time integration. Introducing the global coefficient vectors:

$$\underline{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \vdots \\ \eta_{N_{\text{dof}}}(t) \end{bmatrix}, \quad \underline{q}(t) = \begin{bmatrix} q_1(t) \\ \vdots \\ q_{N_{\text{dof}}}(t) \end{bmatrix}$$

the semi-discrete system takes the compact matrix form:

$$\mathbf{M} \dot{\underline{\eta}}(t) + \mathbf{D} \underline{q}(t) = \underline{0} \quad (2.21)$$

$$\mathbf{M} \dot{\underline{q}}(t) + gH \mathbf{D} \underline{\eta}(t) = \underline{0} \quad (2.22)$$

2.3.5 FINAL BLOCK SYSTEM

Stacking the unknown vectors into the global state vector $\underline{U} = \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix}$, the coupled system (2.21)-(2.22) is cast into the block matrix form:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix} \begin{bmatrix} \dot{\underline{\eta}} \\ \dot{\underline{q}} \end{bmatrix} = - \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \underline{\eta} \\ \underline{q} \end{bmatrix} \quad (2.23)$$

Defining the global block matrices:

$$\mathcal{M} = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{M} \end{bmatrix}, \quad \mathcal{S} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \mathbf{0} \end{bmatrix} \quad (2.24)$$

the target semi-discrete first-order ODE system is recovered:

$$\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \underline{0} \quad (2.25)$$

Here, $\mathcal{M}$ is symmetric and positive definite (diagonal under GLL quadrature), while $\mathcal{S}$ encodes the anti-symmetric spatial derivative operator scaled by the physical parameters.

2.4 MATLAB SOLVER

The semi-discrete block system derived in section 2.3 is advanced in time by means of the Crank-Nicolson scheme. The implementation is validated against the exact traveling-wave solution associated with the Gaussian initial data prescribed by the homework assignment. Accuracy is assessed through direct solution snapshots, h -refinement and p -enrichment convergence studies.

2.4.1 NUMERICAL SETUP AND TIME INTEGRATION

The numerical experiments are performed on the interval $(0, L)$ with

$$L = 1 \text{ m}, \quad H = 1 \text{ m}, \quad g = 9.81 \text{ m/s}^2$$

yielding the wave speed

$$c = \sqrt{gH} = \sqrt{9.81} \text{ m/s} \quad (2.26)$$

and a final time $T = 0.5 \text{ s}$. The initial conditions are the Gaussian pulse prescribed by the assignment:

$$\eta_0(x) = e^{-50(x-0.5)^2}, \quad q_0(x) = ce^{-50(x-0.5)^2} \quad (2.27)$$

Since the datum (2.27) satisfies the compatibility condition $q_0 = c\eta_0$, the exact solution is the periodic right-traveling wave

$$\eta_{\text{ex}}(x, t) = \exp \left[-50 (\text{mod}(x - ct, L) - 0.5)^2 \right] \quad (2.28)$$

Crank-Nicolson discretization The global semi-discrete system reads $\mathcal{M}\dot{\underline{U}} + \mathcal{S}\underline{U} = \underline{0}$. The θ -method with $\theta = \frac{1}{2}$ is adopted, leading to the Crank-Nicolson scheme:

$$\mathcal{M} \frac{\underline{U}^{k+1} - \underline{U}^k}{\Delta t} + \mathcal{S} \left[\frac{1}{2} \underline{U}^{k+1} + \frac{1}{2} \underline{U}^k \right] = \underline{0} \quad (2.29)$$

Collecting all unknowns on the left-hand side, the linear system to be solved at each time step is:

$$\left(\mathcal{M} + \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2}\mathcal{S}\right)\underline{U}^k \quad (2.30)$$

The scheme is second-order accurate in time and unconditionally stable for the present conservative block system. Since the system matrix on the left-hand side of (2.30) depends only on the mesh and on Δt , it is assembled and factorized once before entering the time loop; only the right-hand side is updated at each step.

Discrete L^2 error The accuracy of the numerical approximation is quantified by the L^2 norm of the error on η at the final time:

$$\|\underline{\eta}_{\text{ex}}(\cdot, T) - \underline{\eta}_h(\cdot, T)\|_{L^2(0,L)} = \left(\int_0^L |\underline{\eta}_{\text{ex}}(x, T) - \underline{\eta}_h(x, T)|^2 dx \right)^{1/2} \quad (2.31)$$

Within the finite element framework, this integral is evaluated through the global mass matrix $\mathbf{M}$. Letting $\underline{e} = \underline{\eta}_{\text{ex}}(T) - \underline{\eta}_h(T)$ denote the nodal error vector, the discrete estimate of (2.31) is:

$$E_{L^2} = \sqrt{\underline{e}^T \mathbf{M} \underline{e}} \quad (2.32)$$

To ensure that the exact solution is evaluated at the same time actually reached by the solver, the effective time step is reset to $\Delta t_{\text{eff}} = T/N_{\text{steps}}$ with $N_{\text{steps}} = \text{round}(T/\Delta t)$. This correction prevents a mismatch between the target final time and the numerical time from contaminating the measured convergence rates.

2.4.2 WAVE PROPAGATION SNAPSHOTS

The comparison between the SEM approximation and the exact traveling-wave profile at the final time $T = 0.5$ s is shown in figure 2.1, obtained with $N_e = 20$ elements, polynomial degree $p = 4$ and $\Delta t = 10^{-3}$ s.

Wave packet transport: periodic wrap-around The Gaussian pulse is transported rightward without distortion at the theoretical speed $c = \sqrt{gH}$. The correct enforcement of periodic boundary conditions is evidenced by the re-entry of the pulse from the left boundary after crossing $x = L$. This behavior is reproduced accurately by the SEM solver for both η and q , with no perceptible numerical diffusion.

2.4.3 CONVERGENCE ANALYSIS

The spatial convergence of the scheme is assessed by monitoring the discrete error (2.32) under mesh refinement (h -study) and polynomial enrichment (p -study). The theoretical spatial error estimate for the continuous Galerkin SEM of degree p reads:

$$E_{\text{spatial}} \approx Ch^{p+1} \quad (2.33)$$

provided the exact solution is sufficiently smooth. The total measured error also contains the temporal contribution of the Crank-Nicolson scheme:

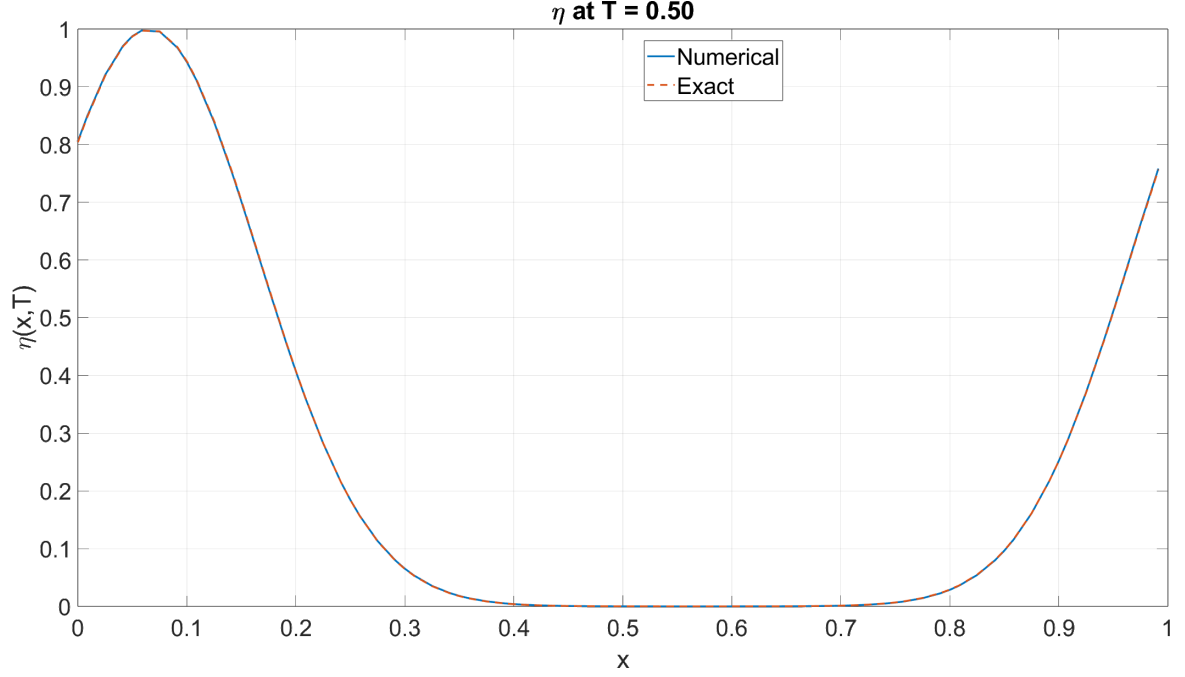
$$E_{\text{total}} \lesssim C_1 h^{p+1} + C_2 \Delta t^2 \quad (2.34)$$

In all convergence tests, Δt is scaled alongside h so that the temporal term remains negligible with respect to the spatial one. The experimental convergence rate between two successive refinements is:

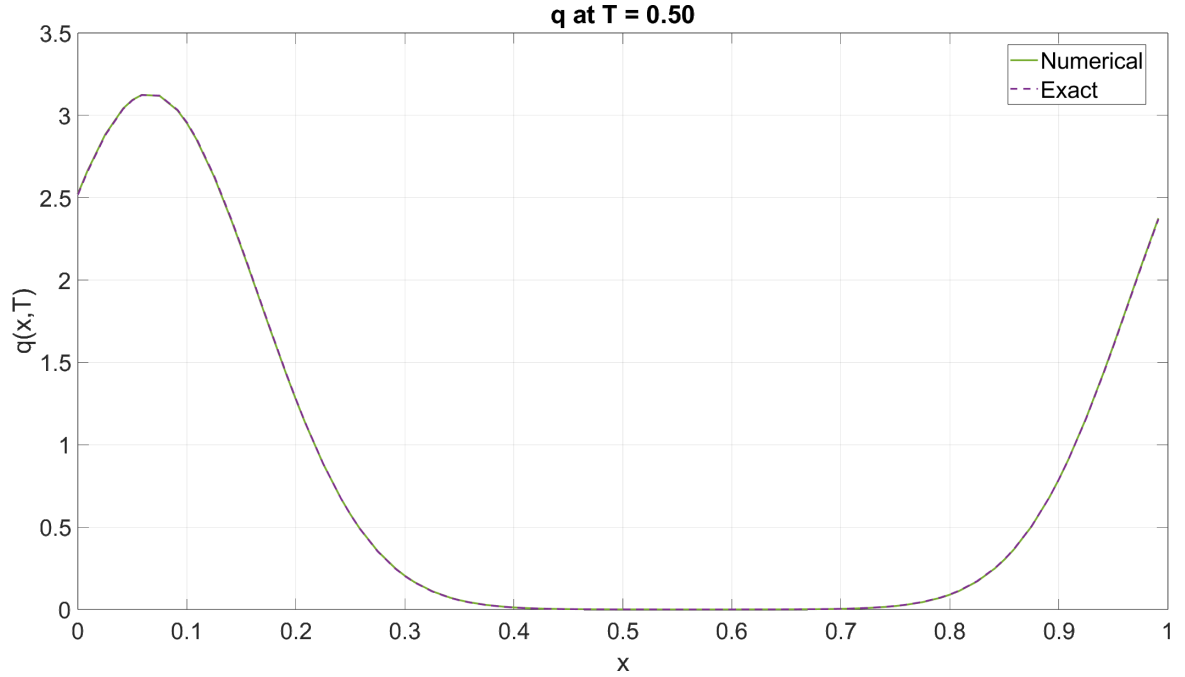
$$r_i = \frac{\log(E_i/E_{i+1})}{\log(h_i/h_{i+1})} \quad (2.35)$$

h -refinement: high-order algebraic convergence The upper panel of figure 2.2 shows that the L^2 error decreases monotonically under mesh refinement, with a slope compatible with the $\mathcal{O}(h^5)$ reference. The slight deviation from the ideal theoretical exponent is physically expected: the Gaussian pulse, when extended periodically through the modulo operation in (2.28), is not globally smooth across the domain boundaries. This limits the regularity assumption underlying estimate (2.33). Furthermore, the second-order temporal error of the Crank-Nicolson scheme contributes an irreducible baseline that prevents the total error from matching the ideal spatial trend exactly.

p -enrichment: monotone decay The lower panel of figure 2.2 confirms that the error decreases monotonically as the polynomial degree p is increased at fixed mesh. The trend is consistent with the high-order accuracy expected from the SEM framework. The same limiting mechanisms identified in the h -study apply here: the limited periodic regularity of the Gaussian datum and the $\mathcal{O}(\Delta t^2)$ temporal floor define the effective accuracy ceiling for any fixed time step. Within these constraints, the observed monotone decay fully validates the spectral element implementation.

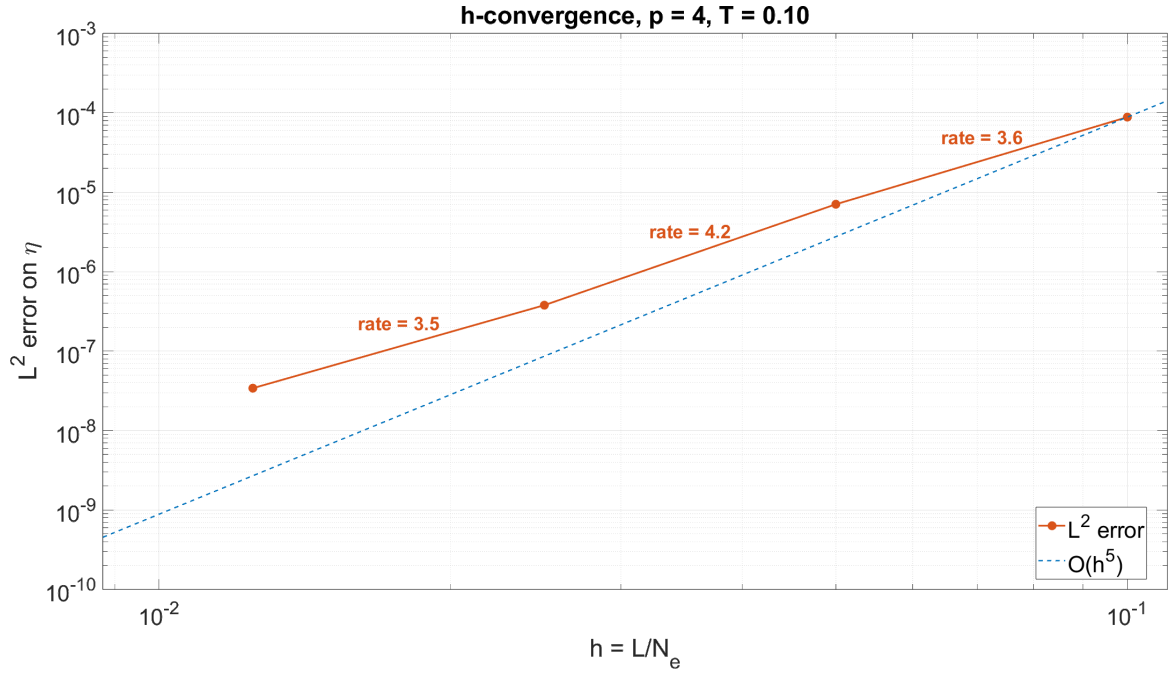


(a) Free-surface elevation $\eta(x, T)$. The numerical profile is visually indistinguishable from the exact solution. Since $cT \approx 1.56 \text{ m} > L$, the pulse has crossed the right boundary and re-entered the domain from the left, providing a direct verification of the periodic connectivity implemented in the solver

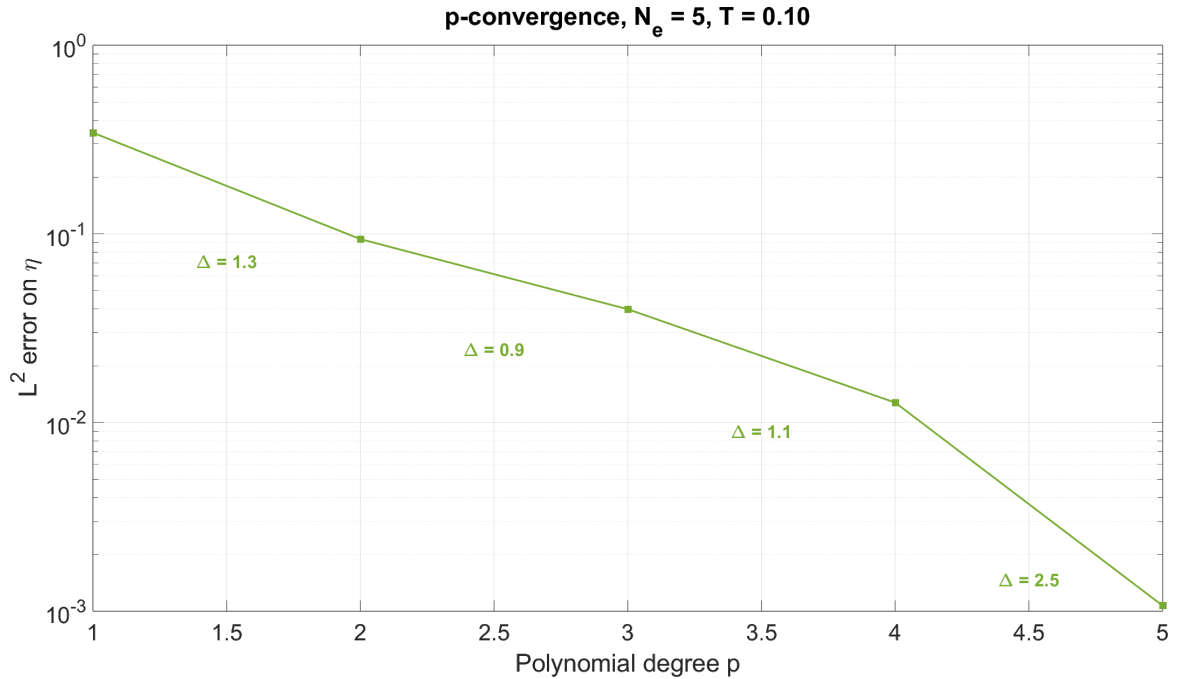


(b) Discharge $q(x, T)$. Since the exact solution satisfies $q_{\text{ex}} = c\eta_{\text{ex}}$ at all times, the discharge reproduces the same translated Gaussian shape. The numerical result captures this proportionality without visible amplitude loss or phase lag, confirming the non-dissipative character of the Crank-Nicolson integrator

Figure 2.1: Wave propagation snapshots for the Gaussian benchmark at final time $T = 0.5 \text{ s}$. Initial conditions as in (2.27), exact solution as in (2.28). Physical parameters: $L = 1 \text{ m}$, $H = 1 \text{ m}$, $g = 9.81 \text{ m/s}^2$. Discretization: $N_e = 20$, $p = 4$, $\Delta t = 10^{-3} \text{ s}$. The wrap-around of the pulse from right to left confirms the correct implementation of periodic BC



(a) Log-log plot of E_{L^2} versus mesh size h at fixed polynomial degree $p = 4$, for meshes $N_e = 10, 20, 40, 80$. The error decreases monotonically and follows a high-order algebraic trend. The local experimental rates are slightly below the ideal exponent of 5, consistently with the limited periodic regularity of the Gaussian datum and the residual second-order temporal error of the Crank-Nicolson integrator



(b) Semi-log plot of E_{L^2} versus polynomial degree p at fixed mesh $N_e = 5$, for $p = 1, 2, 3, 4, 5$. The error decays monotonically as the approximation order increases, reflecting the polynomial enrichment of the SEM space. No visible saturation occurs within the tested range

Figure 2.2: Convergence analysis for the Gaussian benchmark. The L^2 error on η is evaluated at $T = 0.5$ s with the same physical parameters as in figure 2.1. The slight deviation from the ideal spectral slopes is physically expected: the Gaussian profile, when extended periodically through the modulo argument, exhibits limited regularity at the domain boundaries, which weakens the smoothness assumptions underlying (2.33). In addition, the second-order temporal error of the Crank-Nicolson scheme provides a non-negligible lower bound on the total error, as expressed in (2.34). Both effects are consistent with the observed behavior

2.5 STUDIES ON BOUNDARY CONDITIONS AND FRICTION

Two optional modifications of the periodic shallow-water model are examined: the replacement of periodicity by reflecting wall boundary conditions and the introduction of a linear friction term in the momentum equation. In both cases, the theoretical implications for the equivalent second-order equation, the energy balance and the discrete block formulation are derived first. The corresponding numerical evidence is then collected and interpreted in a single dedicated subsection.

2.5.1 PART A: REFLECTING WALL BOUNDARY CONDITIONS

Strong formulation and equivalent second-order condition The periodic boundary conditions are replaced by rigid-wall conditions, namely

$$q(0, t) = 0, \quad q(L, t) = 0 \quad (2.36)$$

These impose zero discharge at both endpoints, describing a closed basin where no flux enters or leaves the domain. The equivalent condition for the second-order wave equation is obtained by evaluating the momentum equation $q_t + gH\eta_x = 0$ at $x = 0$ and $x = L$. Since $q(\cdot, t)$ vanishes at both walls for every t , its time derivative also vanishes there. Therefore:

$$q_t(0, t) = 0 \implies gH\eta_x(0, t) = 0 \implies \eta_x(0, t) = 0 \quad (2.37)$$

and, analogously,

$$q_t(L, t) = 0 \implies gH\eta_x(L, t) = 0 \implies \eta_x(L, t) = 0 \quad (2.38)$$

Hence, homogeneous Dirichlet conditions on q are equivalent to homogeneous Neumann conditions on η . Physically, the wave is reflected by the wall without changing sign.

Effect on the energy balance The conserved energy introduced in section 2.2 is

$$E(t) = \int_0^L \left(\frac{1}{2}g\eta^2 + \frac{1}{2H}q^2 \right) dx \quad (2.39)$$

Repeating the computation carried out for the periodic case yields

$$\frac{dE}{dt} = -g [\eta(x, t)q(x, t)]_{x=0}^{x=L} = -g (\eta(L, t)q(L, t) - \eta(0, t)q(0, t)) \quad (2.40)$$

Substituting (2.36), the boundary contribution vanishes identically:

$$\frac{dE}{dt} = 0 \quad (2.41)$$

Reflecting walls therefore preserve the total mechanical energy exactly. Unlike the periodic setting, energy is no longer transported across the boundary; instead, it remains trapped inside the domain and is redistributed through successive reflections.

Effect on the weak formulation and discrete spaces The function spaces must be adapted. The periodicity constraint is removed, so the free-surface elevation is sought in

$$V = H^1(0, L) \quad (2.42)$$

Since the discharge must satisfy (2.36) strongly, the corresponding trial and test space becomes

$$V_0 = H_0^1(0, L) = \{v \in H^1(0, L) : v(0) = v(L) = 0\} \quad (2.43)$$

The weak problem reads: find $\eta(\cdot, t) \in V$ and $q(\cdot, t) \in V_0$ such that, for all $v \in V$ and $w \in V_0$,

$$\int_0^L \eta_t v dx + \int_0^L q_x v dx = 0 \quad (2.44)$$

$$\int_0^L q_t w dx + gH \int_0^L \eta_x w dx = 0 \quad (2.45)$$

The algebraic block structure remains the same as in section 2.4; only the treatment of the boundary degrees of freedom changes. In practice, the periodic identification of the first and last node is removed and the endpoint degrees of freedom associated with q are constrained to zero.

2.5.2 PART B: LINEAR FRICTION

Modified strong formulation A linear friction term is added to the momentum equation with damping coefficient $\gamma > 0$:

$$\begin{cases} \eta_t + q_x = 0 \\ q_t + gH\eta_x = -\gamma q \end{cases} \quad \gamma > 0 \quad (2.46)$$

The term $-\gamma q$ models a resistive effect that opposes the discharge and progressively removes mechanical energy from the system.

Equivalent damped wave equation The continuity equation is differentiated with respect to time:

$$\eta_{tt} = -q_{xt} \quad (2.47)$$

The momentum equation is differentiated with respect to x :

$$q_{tx} + gH\eta_{xx} = -\gamma q_x \quad (2.48)$$

By Schwarz's theorem, $q_{xt} = q_{tx}$. Using (2.47) and the identity $q_x = -\eta_t$ from the continuity equation:

$$\eta_{tt} + \gamma\eta_t = gH\eta_{xx} \quad (2.49)$$

This is the Telegrapher's equation. The additional first-order time derivative introduces amplitude attenuation while leaving the wave propagation mechanism otherwise unchanged.

Effect on the energy identity Starting from (2.39), the time derivative of the energy is

$$\frac{dE}{dt} = \int_0^L \left(g\eta\eta_t + \frac{1}{H}qq_t \right) dx \quad (2.50)$$

Substituting the modified system (2.46):

$$\frac{dE}{dt} = \int_0^L \left[g\eta(-q_x) + \frac{1}{H}q(-gH\eta_x - \gamma q) \right] dx \quad (2.51)$$

Collecting terms and recognizing the total derivative $\eta q_x + q\eta_x = \frac{d}{dx}(\eta q)$:

$$\frac{dE}{dt} = -g[\eta q]_0^L - \frac{\gamma}{H} \int_0^L q^2 dx \quad (2.52)$$

Under either periodic or reflecting-wall boundary conditions, the boundary term vanishes, yielding:

$$\frac{dE}{dt} = -\frac{\gamma}{H} \int_0^L q^2 dx \leq 0 \quad (2.53)$$

Friction therefore destroys the conservative character of the system: the total energy decays monotonically as long as q is not identically zero.

Effect on the discrete block system At the semi-discrete level, the friction term modifies only the $(2, 2)$ block of the spatial operator. The conservative block matrix $\mathcal{S}$ is replaced by its damped counterpart:

$$\mathcal{S}_{\text{fr}} = \begin{bmatrix} \mathbf{0} & \mathbf{D} \\ gH\mathbf{D} & \gamma\mathbf{M} \end{bmatrix} \quad (2.54)$$

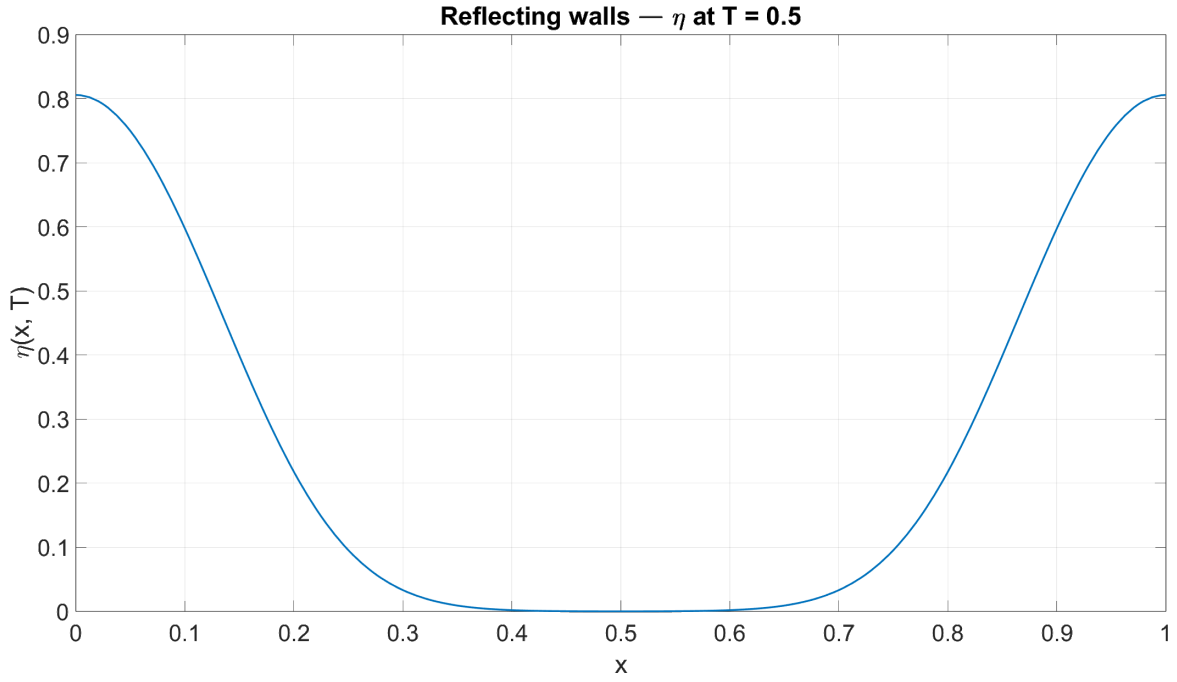
The block mass matrix $\mathcal{M}$ is unchanged. Consequently, the Crank-Nicolson time discretization (2.30) preserves its structure and becomes:

$$\left(\mathcal{M} + \frac{\Delta t}{2} \mathcal{S}_{\text{fr}} \right) \underline{U}^{k+1} = \left(\mathcal{M} - \frac{\Delta t}{2} \mathcal{S}_{\text{fr}} \right) \underline{U}^k \quad (2.55)$$

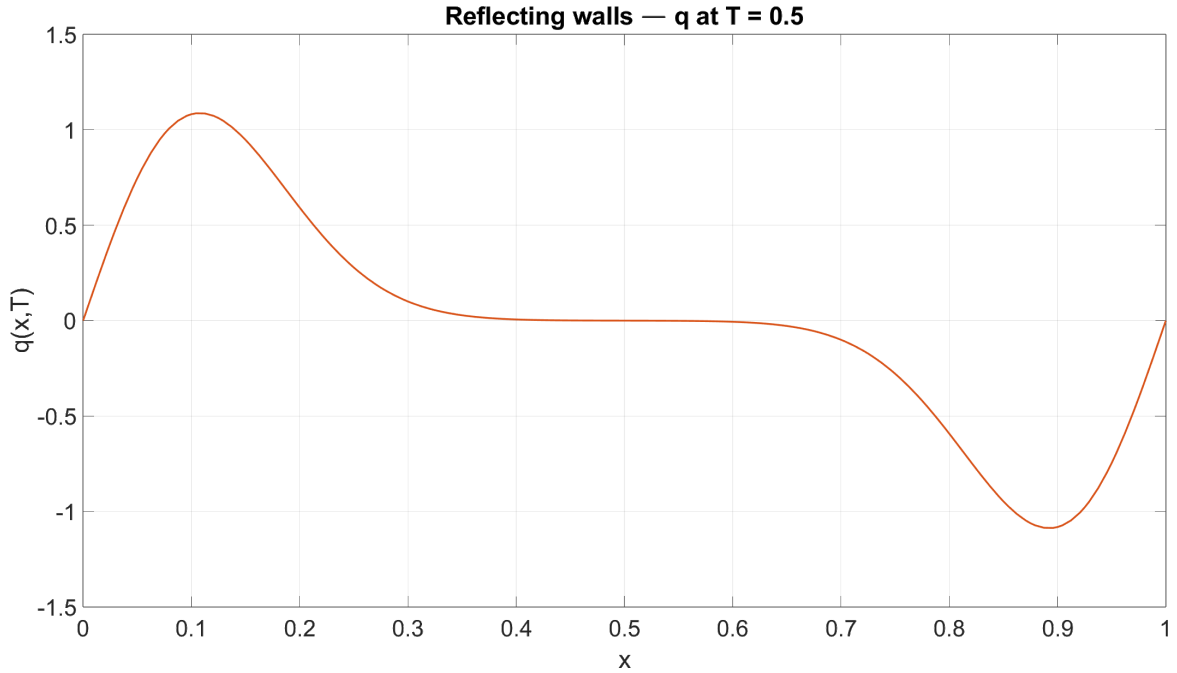
From an implementation standpoint, the friction study requires no new time integrator: it only entails the reassembly of the modified operator (2.54) before entering the time loop.

2.5.3 NUMERICAL RESULTS AND PHYSICAL INTERPRETATION

Two isolated numerical experiments are conducted using the Gaussian initial data (2.27) on the domain $(0, 1 \text{ m})$ with $H = 1 \text{ m}$, $g = 9.81 \text{ m/s}^2$ and final time $T = 0.5 \text{ s}$. The first experiment imposes reflecting walls with $\gamma = 0$; the second retains periodic boundary conditions and activates linear friction with $\gamma = 1$. In both cases, the finest mesh ($N_e = 80$, $p = 4$) and a sufficiently small time step are employed to avoid temporal error contamination. The Gaussian pulse is the most suitable benchmark for these optional studies: its spatial localization makes both the reflection mechanism and the amplitude decay immediately visible in the computed profiles.



(a) Free-surface elevation $\eta(x, T)$ under reflecting-wall boundary conditions (2.36). The initial rightward-propagating Gaussian pulse has reached the right wall and is reflected back into the domain. The reflected profile retains the original amplitude to a very good approximation, in agreement with the conservative energy identity (2.41)



(b) Discharge $q(x, T)$ under reflecting-wall boundary conditions. The discharge goes to exactly zero at both $x = 0$ and $x = L$, enforcing the Dirichlet constraint (2.36). This zero-discharge condition at the walls is the direct mechanism that triggers the conservative in-phase reflection of the free-surface elevation observed in the upper panel

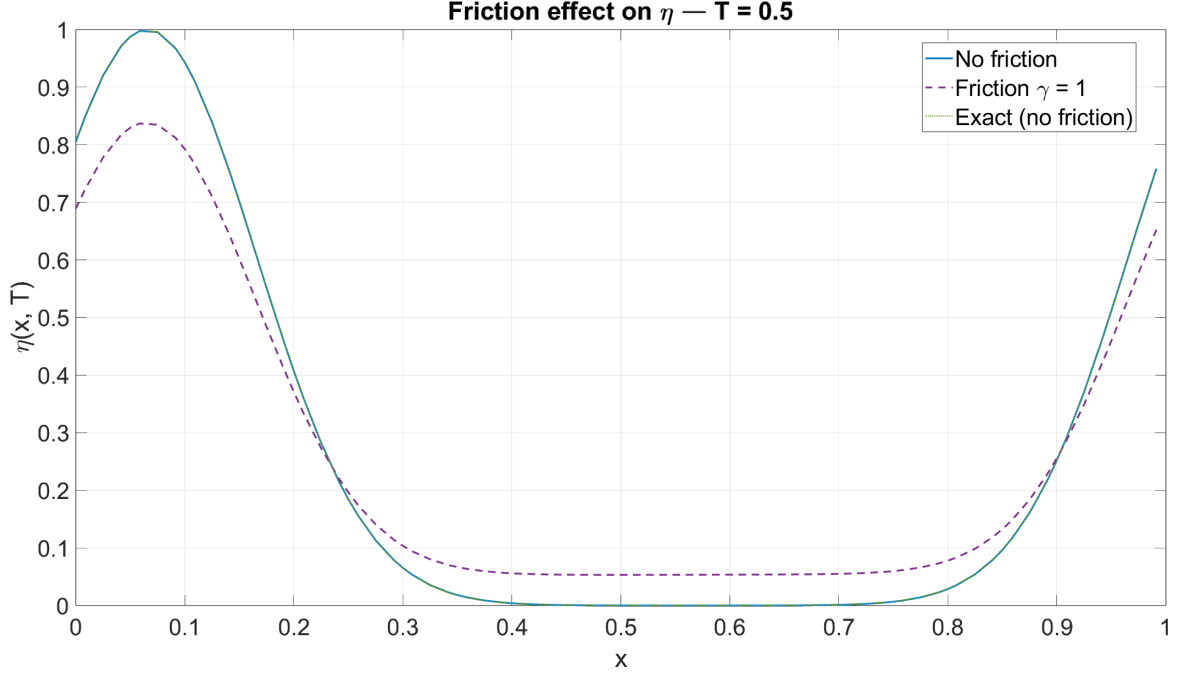
Figure 2.3: Reflecting-wall experiment at $T = 0.5$ s. Physical parameters: $L = 1$ m, $H = 1$ m, $g = 9.81$ m/s², $\gamma = 0$. Discretization: $N_e = 80$, $p = 4$. The reflected wave preserves its amplitude, confirming that $dE/dt = 0$ in a closed domain

Reflecting walls: conservative closed-domain dynamics The discharge profile in figure 2.3(b) shows that q is driven to exactly zero at both endpoints, directly enforcing the boundary condition (2.36). This zero-discharge constraint is the physical mechanism responsible

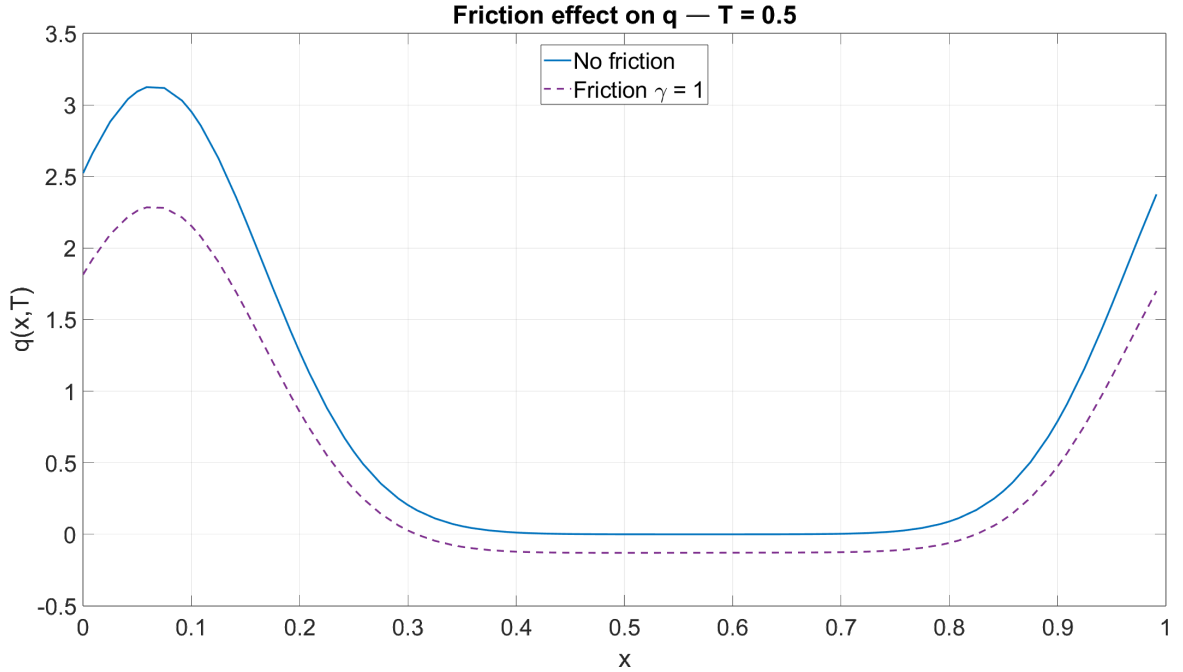
for the in-phase reflection of η : the wave cannot exit the domain in terms of flux, so it reverses direction while retaining its amplitude. The numerical profile in figure 2.3(a) confirms the conservation law (2.41): the reflected pulse carries the same peak height as the incident one and no energy is lost at the wall.

Friction: dissipative transport Figure 2.4 shows that the linear friction term does not alter the propagation speed: the damped profile remains perfectly in phase with the undamped reference. The effect of friction is exclusively dissipative, manifesting as a progressive reduction of the wave amplitude. Both η and q are attenuated by the same factor, consistently with the energy law (2.53). The Telegrapher's equation (2.49) provides the analytical explanation: the additional first-order time derivative $\gamma\eta_t$ introduces exponential amplitude decay while leaving the wave speed c unchanged.

Energy analysis: validation of the analytical predictions Figure 2.5 provides the conclusive numerical validation of the analytical derivations carried out in sections 2.5.1 and 2.5.2. In the frictionless case ($\gamma = 0$), the total discrete energy remains constant throughout the simulation: the flat line confirms that the Crank-Nicolson scheme, combined with the SEM spatial discretization, preserves the conservative energy identity (2.41) to machine precision. This is a non-trivial property: it reflects the symplectic-like character of the Crank-Nicolson integrator when applied to a skew-symmetric spatial operator. When friction is activated ($\gamma = 1$), the energy curve decreases monotonically from its initial value, in full agreement with the theoretical inequality (2.53). The rate of decay is governed by the bulk dissipation term $-(\gamma/H) \int_0^L q^2 dx$, which is largest when the discharge is at its maximum and tapers off as the wave amplitude diminishes over time.



(a) Free-surface elevation $\eta(x, T)$ with linear friction $\gamma = 1$ under periodic boundary conditions. The damped solution (solid line) remains in phase with the undamped reference (dashed), confirming that the wave speed $c = \sqrt{gH}$ is unaffected by the friction term. However, the peak amplitude is significantly reduced, visualizing the monotonic energy dissipation expressed by (2.53)



(b) Discharge $q(x, T)$ with linear friction $\gamma = 1$. The discharge profile undergoes the same amplitude attenuation as the elevation, consistently with the proportionality relation $q_{\text{ex}} = c\eta_{\text{ex}}$ that holds in the undamped limit. The reduction in discharge magnitude is directly responsible for the bulk energy loss through the term $-(\gamma/H) \int_0^L q^2 dx$

Figure 2.4: Friction experiment at $T = 0.5$ s with linear damping coefficient $\gamma = 1$. Physical parameters and discretization as in figure 2.3. The damped wave remains in phase with the undamped reference while exhibiting a clear amplitude reduction, in agreement with the Telegrapher's equation (2.49) and the energy inequality (2.53)

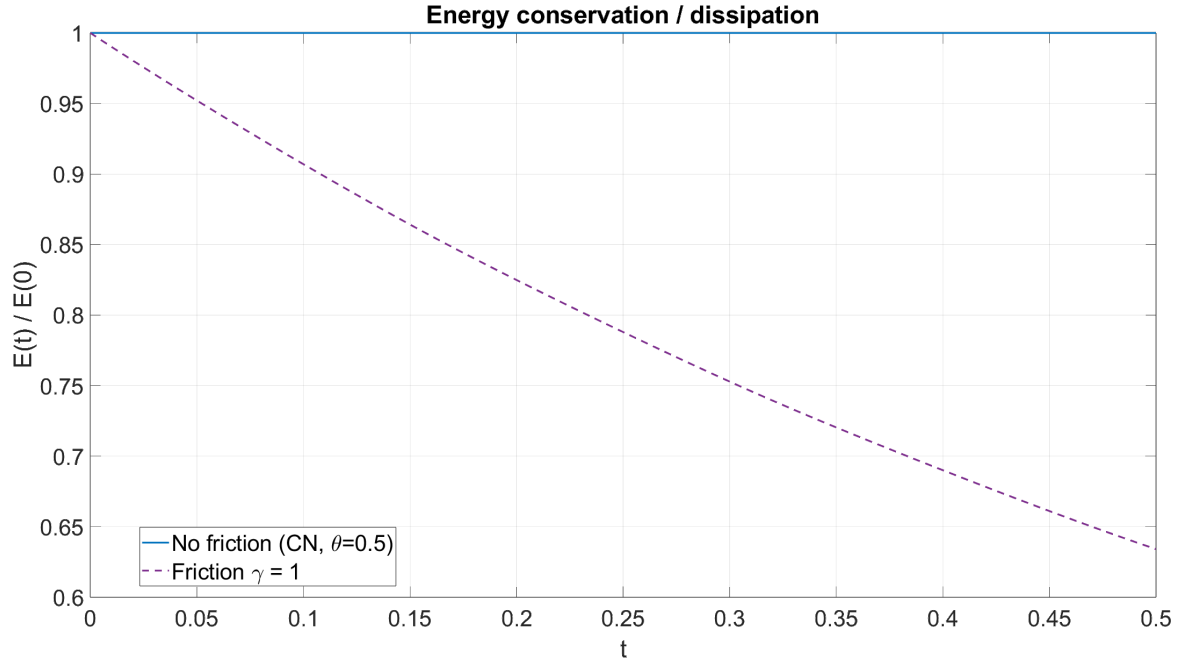


Figure 2.5: Time evolution of the total discrete energy $E(t) = \sum_i (\frac{1}{2}g\eta_i^2 + \frac{1}{2H}q_i^2) \Delta x_i$ for the Gaussian benchmark under two configurations: without friction ($\gamma = 0$, solid line) and with linear friction ($\gamma = 1$, dashed line). Physical parameters and discretization as in figure 2.3. The flat energy curve for $\gamma = 0$ confirms that the combination of the SEM spatial discretization and the Crank-Nicolson time integrator preserves the discrete energy identity (2.41) to machine precision. The monotonically decreasing curve for $\gamma = 1$ is consistent with the dissipation law (2.53), providing quantitative numerical validation of the analytical derivation

3

HOMEWORK 3

3.1 FINITE VOLUME APPROXIMATION

The third homework addresses the numerical solution of the one-dimensional nonlinear Shallow Water Equations (SWE) by means of the Finite Volume Method (FVM), comparing a first-order Godunov scheme and a second-order Lax-Wendroff reconstruction.

3.1.1 THE MATHEMATICAL MODEL

The propagation of water waves in the shallow-water regime is governed, in its full nonlinear formulation, by a system of hyperbolic conservation laws that couples the evolution of the water height $h(x, t)$ and the depth-averaged discharge $q(x, t) = h(x, t)u(x, t)$, where $u(x, t)$ denotes the depth-averaged horizontal velocity. Unlike the linearized shallow water equations treated in the previous chapter, no small-amplitude assumption is invoked here, and the full nonlinear dynamics must be retained. The governing system on the spatial domain $(0, L)$ over the time interval $(0, T]$, with gravitational acceleration g , reads

$$\begin{cases} h_t + (hu)_x = 0 \\ (hu)_t + \left(hu^2 + \frac{1}{2}gh^2\right)_x = 0 \end{cases} \quad (3.1)$$

where the first equation expresses conservation of mass and the second expresses conservation of momentum. Introducing the state vector $\underline{U} = [h, q]^T$ and the flux vector $\underline{F}(\underline{U})$, system (3.1) admits the compact conservative form

$$\underline{U}_t + \underline{F}(\underline{U})_x = 0 \quad (3.2)$$

where the two vectors are explicitly defined as

$$\underline{U} = \begin{bmatrix} h \\ q \end{bmatrix}, \quad \underline{F}(\underline{U}) = \begin{bmatrix} q \\ \frac{q^2}{h} + \frac{1}{2}gh^2 \end{bmatrix} \quad (3.3)$$

The nonlinearity of $\underline{F}$ has a fundamental consequence on the solution structure. The propagation speed of any perturbation is no longer a fixed quantity but depends on the local values of h and u : the two characteristic speeds of system (3.2) are $\lambda_{1,2} = u \mp c$, where $c = \sqrt{gh}$ is the local wave celerity. Because these speeds vary with the solution itself, characteristic curves emanating from different points can intersect in finite time, even when the initial data are smooth. At the crossing point, classical pointwise derivatives cease to exist and a shock, a physical discontinuity in both h and q , forms spontaneously. Classical strong-form

numerical methods based on nodal derivatives (finite differences, finite elements, spectral elements) are therefore inadequate in the vicinity of shocks: they either generate spurious non-physical oscillations or converge to weak solutions that violate the entropy condition and are thus physically incorrect. The conservative form (3.2) provides the correct mathematical framework, since it remains meaningful in the distributional sense across discontinuities and directly encodes the integral conservation laws from which the physical jump conditions (Rankine-Hugoniot relations) are derived.

3.1.2 FVM DISCRETIZATION

The Finite Volume Method is the natural discretization framework for conservation laws in the form (3.2), as it operates directly on integral cell averages rather than pointwise values, thereby preserving conservation at the discrete level and remaining well-defined across discontinuities. The spatial domain $[0, L]$ is partitioned into M uniform cells of width $\Delta x = L/M$. The i -th cell is the interval $[x_{i-1/2}, x_{i+1/2}]$, with center $x_i = (i - \frac{1}{2})\Delta x$ for $i = 1, \dots, M$. The time step Δt is recomputed at each iteration from the CFL stability condition, $\Delta t = \nu \Delta x / \max_i |\lambda|_i$, with Courant number $\nu < 1$, ensuring stability for both schemes. The fundamental discrete unknown is the cell average of the state vector at time t^n :

$$\underline{U}_i^n \approx \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} \underline{U}(x, t^n) dx \quad (3.4)$$

Integrating the conservation law (3.2) over the cell $[x_{i-1/2}, x_{i+1/2}]$ and applying the divergence theorem yields, after division by Δx , the exact integral balance

$$\frac{d}{dt} \underline{U}_i(t) = -\frac{1}{\Delta x} \left[\underline{F}(\underline{U}(x_{i+1/2}, t)) - \underline{F}(\underline{U}(x_{i-1/2}, t)) \right] \quad (3.5)$$

Replacing the continuous interface fluxes with numerical approximations $\underline{F}_{i+1/2}^n$ and applying a forward-Euler time step to (3.5) gives the fully discrete conservative update

$$\underline{U}_i^{n+1} = \underline{U}_i^n - \frac{\Delta t}{\Delta x} \left[\underline{F}_{i+1/2}^n - \underline{F}_{i-1/2}^n \right] \quad (3.6)$$

Equation (3.6) expresses the key property of the FVM: the change in the cell average $\underline{U}_i$ between two time levels depends exclusively on the numerical fluxes at the two bounding interfaces $x_{i\pm 1/2}$. Since every interface flux entering cell i also leaves cell $i+1$ with the opposite sign, the total conserved quantities $\sum_i h_i$ and $\sum_i q_i$ are preserved to machine precision, irrespective of the spatial or temporal resolution. Reflecting wall boundary conditions are enforced at both endpoints through a ghost-cell approach: the ghost cells mirror the water height of the adjacent interior cell while negating the discharge, so that $q = 0$ is imposed at the domain boundaries consistently with the physical requirement of an impenetrable wall.

3.1.3 FIRST-ORDER GODUNOV SCHEME

The Godunov scheme achieves first-order accuracy in both space and time by adopting the simplest consistent reconstruction of the solution within each cell: the state is assumed to be piecewise constant, equal to the cell average $\underline{U}_i^n$ throughout the entire cell $[x_{i-1/2}, x_{i+1/2}]$ at each time level t^n . Under this assumption, the solution profile at any interface $x_{i+1/2}$ exhibits an instantaneous jump from the left state $\underline{U}_i^n$ to the right state $\underline{U}_{i+1}^n$. This configuration constitutes a classical Riemann problem, whose exact solution describes the self-similar wave

pattern consisting of shocks, contact discontinuities, and rarefaction fans that develops instantaneously from the initial discontinuity. The Godunov numerical flux is defined as the exact physical flux evaluated at the solution of this local Riemann problem:

$$\underline{F}_{i+1/2}^{\text{God}} = \underline{F} \left(\underline{U}^* \left(\underline{U}_i^n, \underline{U}_{i+1}^n \right) \right) \quad (3.7)$$

where $\underline{U}^*(\underline{U}_L, \underline{U}_R)$ denotes the state obtained by solving the Riemann problem with left data $\underline{U}_L$ and right data $\underline{U}_R$, evaluated at the interface location $x/t = 0$. Because the Riemann problem is solved exactly, the scheme is upwind-consistent: information is transported in the correct characteristic direction, and the method automatically selects the entropy-satisfying solution even in the presence of sonic rarefactions and strong shocks. The scheme is therefore exceptionally robust and produces monotone, non-oscillatory profiles in the vicinity of discontinuities, correctly capturing both shock speeds and post-shock states. Its primary drawback is the excessive numerical dissipation inherent to the piecewise-constant reconstruction: smooth regions of the solution are smeared over $\mathcal{O}(\Delta x)$ cells, and rarefaction fans appear artificially broadened.

3.1.4 SECOND-ORDER LAX-WENDROFF RECONSTRUCTION

The dissipation inherent to the Godunov scheme can be substantially reduced by improving the spatial accuracy of the reconstruction from first to second order. Instead of assuming a constant state within each cell, a piecewise linear reconstruction is adopted: the solution inside each cell is represented as a linear polynomial whose slope is inferred from the difference between neighboring cell averages. The reconstructed values at the left and right sides of the interface $x_{i+1/2}$, denoted $\underline{U}_{i+1/2}^-$ (approaching from cell i) and $\underline{U}_{i+1/2}^+$ (approaching from cell $i+1$), are obtained by extrapolating the respective cell averages to the interface:

$$\underline{U}_{i+1/2}^- = \underline{U}_i + \frac{1}{2} (\underline{U}_{i+1} - \underline{U}_i) \quad (3.8)$$

$$\underline{U}_{i+1/2}^+ = \underline{U}_{i+1} - \frac{1}{2} (\underline{U}_{i+1} - \underline{U}_i) \quad (3.9)$$

In the limit $\underline{U}_{i+1} \rightarrow \underline{U}_i$, both reconstructed states reduce to the common cell value, recovering the Godunov piecewise-constant reconstruction as expected. For a general jump, the two reconstructed states are passed to the Riemann solver in place of the raw cell averages, yielding the second-order numerical flux

$$\underline{F}_{i+1/2}^{\text{LW}} = \underline{F}^* \left(\underline{U}_{i+1/2}^-, \underline{U}_{i+1/2}^+ \right) \quad (3.10)$$

where $\underline{F}^*$ denotes the numerical flux returned by the Riemann solver evaluated at the reconstructed interface states. The piecewise-linear reconstruction reduces the truncation error from $\mathcal{O}(\Delta x)$ to $\mathcal{O}(\Delta x^2)$ in smooth regions, yielding significantly sharper resolution of both smooth waves and post-shock profiles. However, the higher-order reconstruction introduces a dispersive error component: near sharp discontinuities, where the linear slope approximation is not bounded, the method may generate spurious oscillations (Gibbs-like ripples) that propagate away from the shock front. This trade-off between accuracy in smooth regions and spurious oscillations at discontinuities motivates the introduction of slope limiters such as the minmod or van Leer limiters, which are however beyond the scope of the present work.

3.2 RIEMANN PROBLEMS AND PHYSICAL INTERPRETATION

All three Riemann problems are solved on the domain $(0, L)$ with $L = 1$ m, $g = 9.81$ m/s² and final time $T = 1$ s, employing a uniform FVM mesh. In all three test cases the discharge q is driven to zero at both $x = 0$ and $x = L$ at every time level, confirming the correct implementation of the reflecting wall boundary conditions enforced through the ghost-cell approach described in section 3.1.2.

3.2.1 DAM BREAK AND REVERSE DAM BREAK

Dam break: physical structure The dam-break problem is initialized with water height $h_L = 1$ on the left half of the domain and $h_R = 0.1$ on the right. The initial discontinuity at $x = L/2$ immediately generates two wave families that propagate in opposite directions. A smooth rarefaction fan travels leftward, continuously expanding the region of intermediate height and gradually accelerating the fluid. A shock wave travels rightward as a steep, near-vertical front, carrying the abrupt transition between the intermediate state and the undisturbed low water on the right. The discharge q is positive throughout the domain, reflecting the net mass transport from the high reservoir towards the low region. After reaching the right wall, the shock is reflected and the discharge drops back to zero at the boundary, as seen in figure 3.1.

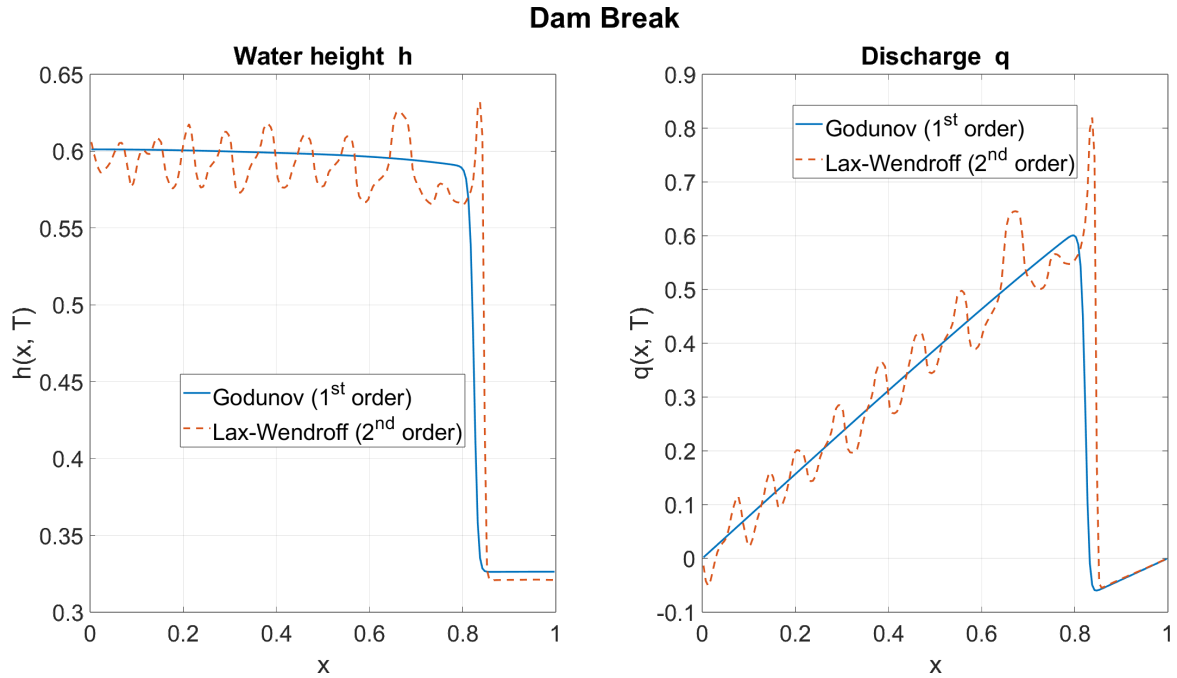


Figure 3.1: Dam-break Riemann problem at $T = 1$ s. Left panel: water height $h(x, T)$ for both schemes. The Godunov scheme (solid blue) captures the rarefaction and the shock monotonically but with visibly diffused fronts; the Lax-Wendroff reconstruction (dashed orange) sharpens both features at the cost of pronounced spurious oscillations in the plateau region and at the shock. Right panel: discharge $q(x, T)$; the positive values confirm a uniform rightward mass transport, while $q = 0$ at both walls verifies the reflecting boundary conditions. Physical parameters: $L = 1$ m, $g = 9.81$ m/s², $h_L = 1$, $h_R = 0.1$

Dam break: numerical comparison The Godunov scheme captures the shock in a rigorously monotone fashion: no overshoot or undershoot appears, and the solution satisfies the

entropy condition by construction. The price paid is significant numerical diffusion — the shock front, which is a true mathematical discontinuity, is spread over several mesh cells, and the rarefaction fan is similarly smeared. The Lax-Wendroff reconstruction resolves both the rarefaction and the shock with considerably higher spatial sharpness, reducing the transition width compared to Godunov. However, the absence of a slope limiter allows unphysical oscillations to develop: the plateau region between the rarefaction tail and the shock front is corrupted by persistent Gibbs-like ripples in both h and q , and a pronounced overshoot appears immediately upstream of the shock front.

Reverse dam break: mirror dynamics The reverse dam-break configuration, initialized with $h_L = 0.1$ and $h_R = 1$, produces a wave structure that is the spatial mirror image of the standard dam break. The shock now travels leftward from the initial discontinuity, while the rarefaction fan propagates to the right. The discharge q is uniformly negative, indicating mass transport from right to left, and reaches zero at both walls as required by the boundary conditions. The same qualitative contrast between the two schemes is observed as in the forward case, confirming that the numerical properties of the Godunov and Lax-Wendroff methods are symmetric with respect to the direction of propagation.

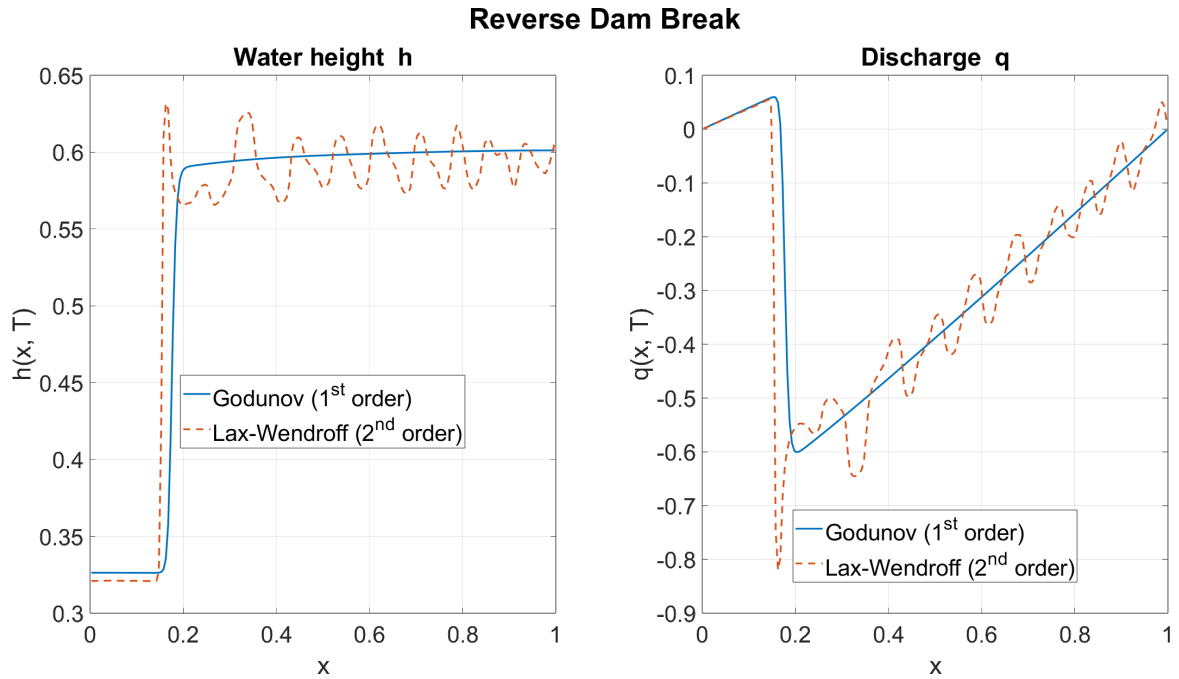


Figure 3.2: Reverse dam-break Riemann problem at $T = 1$ s. Left panel: water height $h(x, T)$. The shock propagates leftward, producing a sharp front near $x \approx 0.15$; the Lax-Wendroff reconstruction (dashed orange) introduces Gibbs oscillations on the high-water plateau to the right of the shock, while the Godunov scheme (solid blue) remains monotone. Right panel: discharge $q(x, T)$; negative values throughout confirm the leftward mass transport, with $q = 0$ enforced at both domain boundaries. Physical parameters: $L = 1$ m, $g = 9.81$ m/s², $h_L = 0.1$, $h_R = 1$

3.2.2 STATIONARY SHOCK (HYDRAULIC JUMP)

Physical structure of the hydraulic jump The third test case is initialized with a height ratio h_L/h_R and initial momentum q chosen so that the Rankine-Hugoniot conditions yield a shock speed of approximately zero. The result is a stationary hydraulic jump, an abrupt

transition in water height that remains near-fixed at the centre of the domain over the entire simulation interval. On the left side of the jump, the water is shallow and fast (supercritical flow, Froude number $Fr > 1$); on the right, it is deep and slow (subcritical flow, $Fr < 1$). The discharge profile transitions smoothly from the left-wall value of zero, rises through the supercritical region, and drops sharply across the jump, correctly returning to zero at the right wall in accordance with the reflecting boundary conditions.

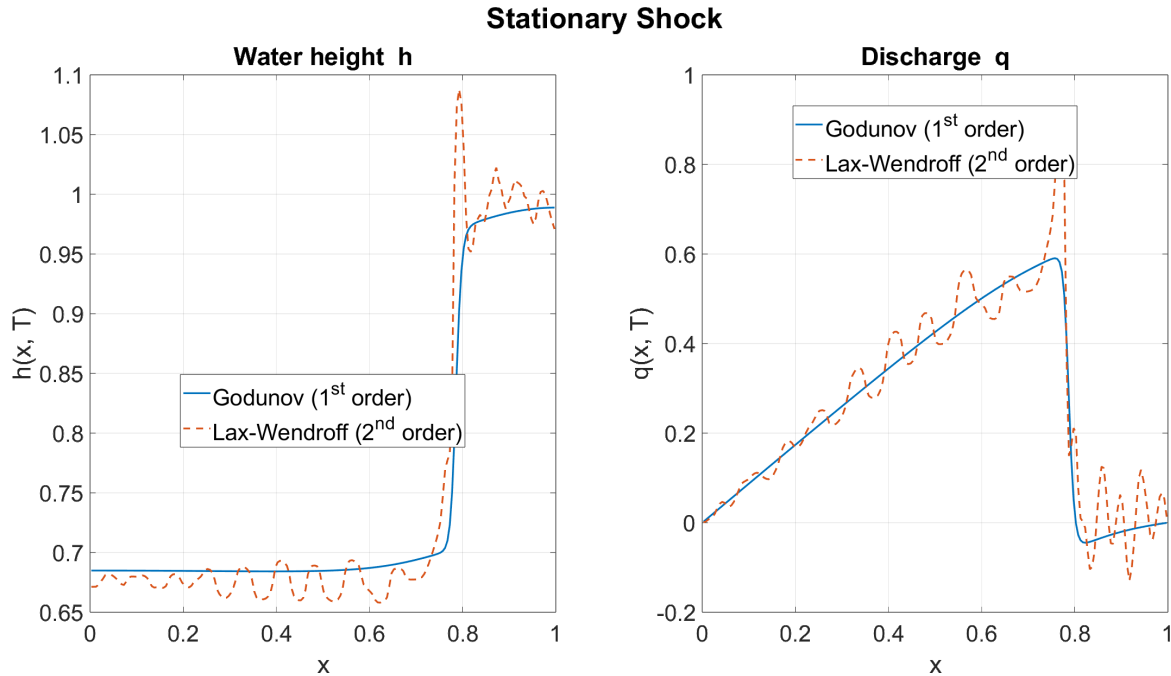


Figure 3.3: Stationary-shock Riemann problem at $T = 1$ s. Left panel: water height $h(x, T)$. The hydraulic jump is located near $x \approx 0.6$; the Godunov scheme (solid blue) resolves it as a smooth, diffused transition, whereas the Lax-Wendroff reconstruction (dashed orange) produces a sharp but heavily oscillating profile, with overshoots clearly visible above the post-shock height. Right panel: discharge $q(x, T)$; the near-zero values on the left of the jump confirm the supercritical regime, while the strong Gibbs ripples of the Lax-Wendroff solution are most pronounced in the vicinity of the discontinuity. Physical parameters: $L = 1$ m, $g = 9.81$ m/s², reflecting walls at both ends

Amplified dispersive errors and the need for slope limiters The stationary shock constitutes the most demanding test for the Lax-Wendroff reconstruction. Because the jump is fixed in space rather than advecting across the mesh, the dispersive error accumulates coherently over the entire time interval instead of being swept away. As a consequence, the Gibbs oscillations visible in figures 3.1 and 3.2 become substantially more pronounced here: the dashed-orange curve in figure 3.3 exhibits large-amplitude, high-frequency ripples that extend well into both the subcritical and supercritical regions. The Godunov scheme, by contrast, remains entirely monotone and converges to the correct shock position, confirming the theoretical robustness discussed in section 3.1.3. The comparison between the two schemes in figure 3.3 provides clear visual evidence that second-order reconstruction without limiting is unsuitable for problems containing strong stationary discontinuities. The natural remedy is the introduction of Total Variation Diminishing (TVD) slope limiters, such as the min-mod or van Leer limiters, which adaptively suppress the linear reconstruction in the vicinity of shocks while retaining second-order accuracy in smooth regions, thereby combining the monotonicity of the Godunov scheme with the sharper resolution of Lax-Wendroff.

3.3 MODIFIED PRESSURE FLUX

The standard SWE system (3.2) assumes a hydrostatic pressure law of the form $p(h) = \frac{1}{2}gh^2$, which leads to the quadratic pressure term in the second component of the flux vector (3.3). This section considers a modification in which the pressure law is replaced by a logarithmic form, giving rise to an alternative conservative system whose mathematical structure and wave dynamics differ in a physically meaningful way from the standard model.

3.3.1 ANALYTICAL FORMULATION

The modified pressure law is defined as

$$p(h) = gh \log \left(\frac{h}{h_{\text{ref}}} \right) \quad (3.11)$$

where $h_{\text{ref}} > 0$ is a fixed reference height. Substituting (3.11) into the momentum equation in place of the standard hydrostatic term yields the modified flux vector

$$\underline{F}(\underline{U}) = \begin{bmatrix} q \\ \frac{q^2}{h} + gh \log \left(\frac{h}{h_{\text{ref}}} \right) \end{bmatrix} \quad (3.12)$$

while the mass conservation equation and the state vector $\underline{U} = [h, q]^T$ remain unchanged. The system retains the conservative form (3.2) with the flux (3.12), and the same FVM update formula (3.6) applies without modification: only the evaluation of the flux vector at each interface changes. The mathematical properties of the modified system differ from those of the standard SWE because the Jacobian $\mathbf{J} = \partial \underline{F} / \partial \underline{U}$ now takes the form

$$\mathbf{J}(\underline{U}) = \begin{bmatrix} 0 & 1 \\ -\frac{q^2}{h^2} + g \log \left(\frac{h}{h_{\text{ref}}} \right) + g & \frac{2q}{h} \end{bmatrix} \quad (3.13)$$

The eigenvalues of $\mathbf{J}$, which govern the characteristic speeds of the system, are

$$\lambda_{1,2} = u \mp c_{\text{mod}}, \quad c_{\text{mod}} = \sqrt{g \left[\log \left(\frac{h}{h_{\text{ref}}} \right) + 1 \right]} \quad (3.14)$$

where c_{mod} plays the role of the wave celerity for the modified system. Comparing with the standard celerity $c = \sqrt{gh}$, the modified celerity c_{mod} depends logarithmically rather than algebraically on h , which has several physical consequences. For $h > h_{\text{ref}}e^{-1}$ the argument of the square root in (3.14) is positive and the system remains hyperbolic; below this threshold the characteristic speeds become complex, signaling a loss of hyperbolicity that renders the initial-value problem ill-posed. In the hyperbolic regime, the slower growth of c_{mod} with h compared to $\sqrt{gh}$ implies that the modified system propagates waves more slowly at large depths and produces weaker shock strengths for the same initial height discontinuity. The rarefaction fans are correspondingly broader, and the overall flow is less compressive than in the standard shallow-water setting. These differences in characteristic structure are directly observable in the numerical solutions discussed below.

3.3.2 NUMERICAL RESULTS AND DISCUSSION

The modified flux (3.12) is tested with $h_{\text{ref}} = 1$ on the dam-break and stationary-shock initial data, using the first-order Godunov scheme for both the standard and modified systems to isolate the effect of the pressure law from any scheme-dependent dissipation or dispersion.

Dam break with modified pressure Figure 3.4 compares the two Godunov solutions at time $T = 1$ s for the dam-break initial data ($h_L = 1$, $h_R = 0.1$). The most immediate effect of the modified pressure is a marked reduction in the rightward shock speed: at $T = 1$ the shock front computed with the standard flux has already reached and reflected from the right wall, whereas the modified-pressure shock is still propagating through the interior of the domain, having traveled a substantially shorter distance. This behavior is in direct agreement with the eigenvalue analysis of section 3.3.1: for $h \leq 1$ and $h_{\text{ref}} = 1$ one has $\log(h/h_{\text{ref}}) \leq 0$, so that $c_{\text{mod}} \leq \sqrt{g}$ is strictly smaller than the corresponding standard celerity $c = \sqrt{gh}$ evaluated at the same depth. The weaker pressure gradient of the logarithmic law provides less driving force per unit height, resulting in a slower propagation of the shock front and a less pronounced post-shock height plateau. The rarefaction fan on the left is also affected: the modified solution shows a smoother and more extended depletion profile rather than the flat constant-state region produced by the standard flux, reflecting the altered characteristic fan structure. The discharge panel confirms both observations: the peak value of q is significantly lower in the modified case, and the profile shape is qualitatively different, with a more gradual spatial variation across the entire domain.

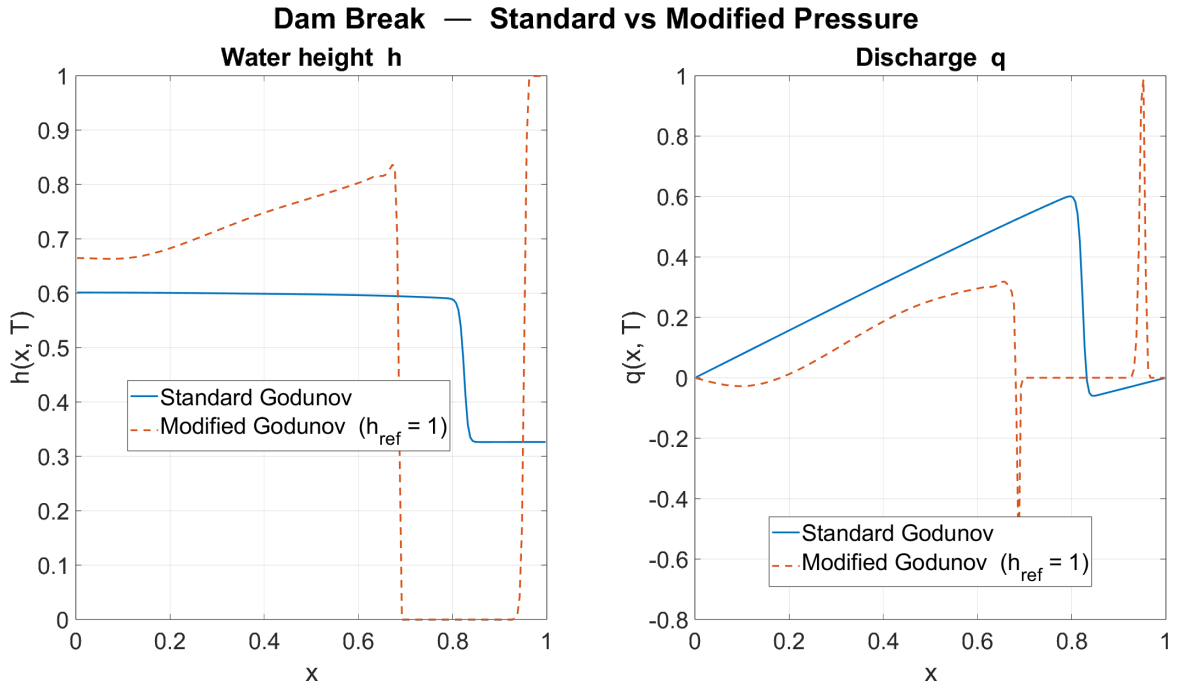


Figure 3.4: Dam-break Riemann problem at $T = 1$ s: comparison between the standard Godunov scheme (solid blue) and the modified-pressure Godunov scheme with $h_{\text{ref}} = 1$ (dashed orange). Left panel: water height $h(x, T)$. The modified shock front has not yet reached the right wall, travelling at a lower speed consistent with the reduced wave celerity c_{mod} predicted by (3.14); the post-shock plateau is lower and the rarefaction profile is more gradual. Right panel: discharge $q(x, T)$; the peak discharge is substantially reduced in the modified case, reflecting the weaker momentum transfer induced by the logarithmic pressure term. Physical parameters: $L = 1$ m, $g = 9.81$ m/s², $h_L = 1$, $h_R = 0.1$, $h_{\text{ref}} = 1$

Stationary shock with modified pressure The most striking consequence of the pressure modification is revealed by the stationary-shock test, shown in figure 3.5. In the standard system, the initial data (h_L, h_R, q) are constructed precisely so that the Rankine-Hugoniot jump condition yields a shock speed $\sigma = 0$: the net flux imbalance across the discontinuity vanishes, $[F_{\text{std}}] = 0$, and the shock remains stationary at the center of the domain for all time. Replacing the standard flux with the modified flux (3.12) alters the second component of the flux vector at every height level, so that the same left and right states now produce a non-zero flux jump $[F_{\text{mod}}] \neq 0$. The Rankine-Hugoniot condition for the modified system,

$$\sigma[U] = [F_{\text{mod}}(U)] \quad (3.15)$$

is therefore satisfied only for a strictly positive shock speed $\sigma > 0$, meaning that the hydraulic jump is no longer stationary but drifts rightward at a finite velocity. This is precisely what figure 3.5 shows: at $T = 1$ s the modified-pressure shock has migrated visibly to the right of the standard one, while both solutions retain a sharp, monotone profile characteristic of the first-order Godunov scheme. The pre-shock heights on the left plateau also differ slightly between the two cases, reflecting the different intermediate states produced by the respective Riemann solvers. The discharge panel corroborates the picture: the modified solution exhibits a higher and spatially shifted peak, consistent with a shock that is still actively propagating and carrying momentum to the right rather than being in a force-balanced stationary configuration.

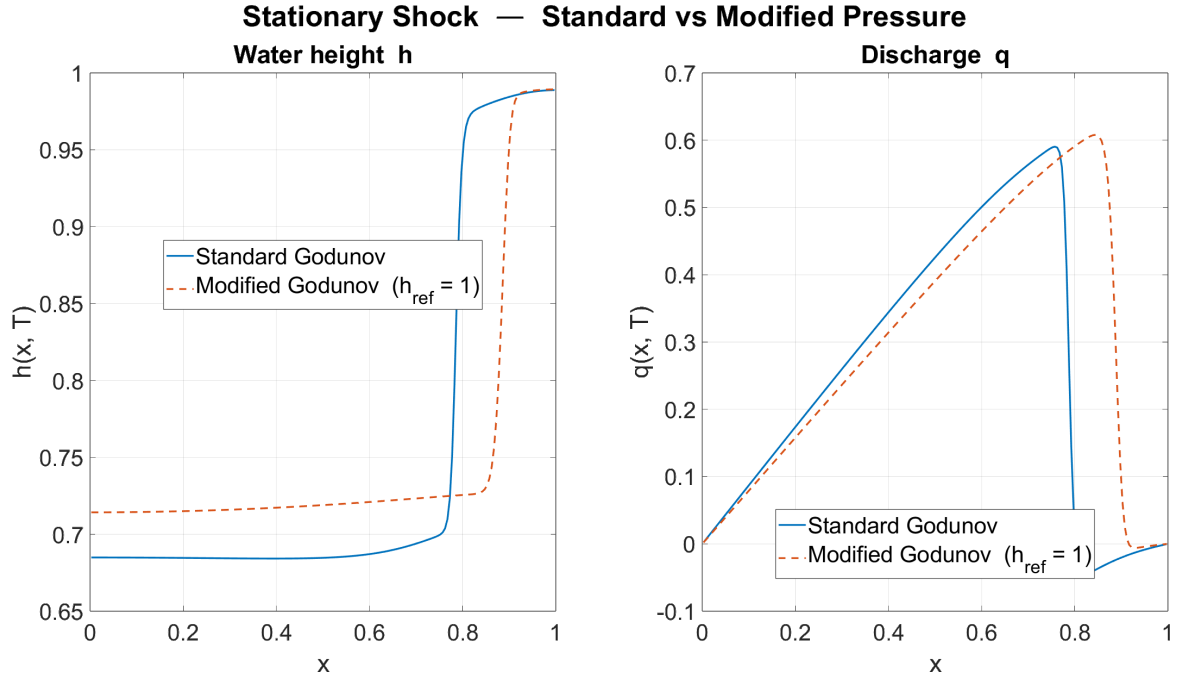


Figure 3.5: Stationary-shock Riemann problem at $T = 1$ s: comparison between the standard Godunov scheme (solid blue) and the modified-pressure Godunov scheme with $h_{\text{ref}} = 1$ (dashed orange). Left panel: water height $h(x, T)$. The initial data were tuned so that the standard Rankine-Hugoniot condition yields $\sigma = 0$; the modified pressure alters the flux balance according to (3.15), producing a positive shock speed that causes the hydraulic jump to drift rightward, as clearly visible in the shifted front position. Right panel: discharge $q(x, T)$; the higher peak and rightward displacement of the modified solution confirm the active momentum transport induced by the non-zero shock speed. Physical parameters: $L = 1$ m, $g = 9.81$ m/s², $h_L = 1$, $h_R = 0.5$, $h_{\text{ref}} = 1$